Stability and Collapse of Pollination Networks:
A Network Science Perspective

Network Science Project Report
Indian Institute of Technology Bhilai

Authors:
Agastya Nath | Aman Pratap Singh | Dhananjoy Doley | Paritosh Lahre | Rahul Dev Reddy
April 2026

Abstract
Pollination networks represent among the most ecologically critical mutualistic systems on Earth, underpinning the reproduction of approximately 88% of flowering plant species and sustaining nearly 75% of global food crops. Yet these networks face mounting pressure from habitat loss, pesticide use, climate change, and invasive species, driving rapid pollinator declines worldwide. This project applies a unified network science framework to characterize the structure, resilience, and collapse dynamics of pollination networks across multiple scales — from individual datasets to continent-level aggregations spanning 172 real-world bipartite interaction networks. Our methodology integrates classical network metrics (degree, betweenness, and eigenvector centrality; modularity; nestedness), cascading co-extinction simulation, climate change stress modelling, adaptive interaction rewiring, early warning signal detection, machine learning extinction risk prediction, Graph Neural Network (GNN)-based functional role discovery, and a novel species transplantation analysis. Key findings include: (i) continent-level networks display markedly different structural signatures, with South America showing the highest modularity (Q = 0.781) and North America the most vulnerable topology under targeted extinction; (ii) a GNN framework applied across 172 networks achieves a mean link-prediction AUC of 0.960 and reveals a connectance-driven gradient from Generalist-dominated small networks to Specialist/Connector-dominated large ones; (iii) species transplantation of a pollinator into a target community increases interaction count while marginally reducing nestedness, demonstrating measurable but network-dependent integration; and (iv) eigenvector centrality is the strongest predictor of extinction risk across all datasets. Collectively, these results provide a data-driven, multi-methodological assessment of pollination network fragility and offer actionable insights for ecosystem conservation.

1. Introduction and Motivation

Pollination is one of the most important ecological processes sustaining life on Earth. The transfer of pollen between flowering plants — mediated by insects, birds, bats, and other animals — drives the reproduction of a staggering proportion of plant diversity. Approximately 87.5% of flowering plant species depend on animal pollinators, and nearly 75% of globally important food crops rely on animal-mediated pollination. The economic value of pollination services to agriculture has been estimated in the hundreds of billions of dollars annually, making the health of pollinator communities a matter of both ecological and economic urgency.

Despite this importance, pollinator populations worldwide are declining at alarming rates. Large-scale surveys document the collapse of bee populations in North America and Europe, the disappearance of specialist pollinators from degraded habitats, and the phenological mismatches induced by climate change that desynchronize plants from their pollinators. These declines are driven by a confluence of anthropogenic stressors: pesticide exposure (particularly neonicotinoids), habitat fragmentation, introduction of invasive species and pathogens, land-use change, and the accelerating effects of climate warming. Understanding how

these stressors interact with the underlying structure of pollination networks is essential for predicting and preventing cascading biodiversity loss.

Traditional ecological approaches to studying pollination focus on pairwise species interactions or on the abundance of individual pollinator taxa. While informative, such approaches fail to capture the systemic, interconnected nature of pollination communities. A single network-level perturbation — the loss of a highly connected generalist pollinator, for instance — can trigger cascading extinctions affecting dozens of plant species and the other pollinators that depend on them. These emergent, higher-order dynamics are invisible to pairwise analyses but readily apparent when pollination is modelled as a complex network.

Network science provides a natural and powerful toolkit for studying these interactions. By representing pollinators and plants as nodes and their interactions as edges in a bipartite graph, we can compute centrality measures that identify critical species, detect community structure (modularity) that governs how perturbations propagate, quantify nestedness patterns that influence resilience, and simulate cascading failures under diverse extinction scenarios. More recently, Graph Neural Networks (GNNs) have emerged as a data-driven alternative to hand-crafted ecological metrics, capable of learning latent representations of network structure that reveal functional roles beyond what classical indices capture.

Our study spans six continents, 172 individual networks from the Web of Life database, and introduces a few methodological novelties including climate stress simulation, adaptive rewiring, early warning signal detection, ML-based extinction risk ranking, GNN-based role discovery, and species transplantation impact modelling.

2. Related Literature

The study of pollination networks relies on both ecological theory and advanced complex systems methodologies to understand ecosystem structure, resilience, and collapse dynamics. Here we review the literature that directly informs our analytical framework.

The foundational observation that plant-animal mutualistic networks exhibit a characteristic nested architecture was established by Bascompte et al. (2003). Rohr et al. (2014) formalised how this structural stability relates to environmental perturbations, directly informing our climate stress simulations. Crucially, Lever et al. (2014) showed that while highly nested networks are resilient to gradual changes, they can collapse suddenly and synchronously once a critical threshold is crossed—a finding that directly motivates our early warning signal analysis.

Memmott et al. (2004) first demonstrated that pollination networks are robust to random species removal but highly vulnerable to the targeted removal of connected hubs, forming the basis of our cascading co-extinction simulations. More recently, Bascompte and Scheffer (2023) synthesized these concepts into a unified framework for the resilience of plant-pollinator networks, emphasizing how structural stability and interaction rewiring mitigate the impacts of global environmental change. To predict potential collapses under these changes, we operationalise the statistical indicators of "critical slowing down" described by Dakos and Bascompte (2014).

Our project operationalises biodiversity-ecosystem function relationships at a multi-network scale. Simpson et al. (2022) provided a vital empirical demonstration that the number of functionally important bee species scales up with plant community richness. Furthermore, they found that up to 25% of functionally important pollinator species are locally rare. This corroborates our multi-scale findings regarding specialist species and proves that restricting conservation focus strictly to abundant generalists risks missing structurally critical rare pollinators.

Network science provides a unifying framework for understanding dynamic interactions across vastly different systems. For instance, Ward (2022) demonstrated how network engagement, homophily, and local neighborhood choices drive dynamic adaptation and participation within human social networks. In an ecological context, this principle of structural adaptability is mirrored by interaction turnover. CaraDonna et al. (2017) and Valdovinos et al. (2013) provide the empirical and theoretical grounding for our adaptive rewiring simulation, proving that rapid interaction rewiring naturally extends the persistence of mutualistic networks after species loss. Finally, to systematically discover these functional roles without relying on hand-crafted metrics, we utilize the GraphSAGE framework introduced by Hamilton et al. (2017) to generalise node embeddings through multi-hop neighbourhood aggregation.

3. Methods and Methodology

3.1 Datasets

To support our various analytical modules, we utilized a single, comprehensive dataset collection comprising 174 local, city-level plant-pollinator bipartite networks sourced from the Web of Life database. To analyze macro-ecological patterns and stability, these 174 individual city-level datasets were merged according to their geographical coordinates into six aggregated continental networks: Africa, Asia, Europe, North America, Oceania, and South America. In contrast to the continental aggregations used for the stability analysis, the Graph Neural Network (GNN) functional role discovery was performed strictly at the local, city level. For this module, we utilized the full corpus of 174 individual plant-pollinator bipartite networks without applying any geographical merging.

3.1.1 Continent-Level Aggregated Networks

For the primary stability and collapse analysis, we compiled aggregated bipartite pollination networks for six continental regions: Africa, Asia, Europe, North America, Oceania, and South America. These were sourced from the Web of Life database (www.web-of-life.es) and publicly available ecological network repositories. Each continental dataset represents interactions between plant species and diverse pollinator guilds (including hummingbirds, insects, and other arthropods). Network sizes ranged from 127 plants and 299 pollinators (Oceania) to 543 plants and 2,947 pollinators (Asia). All networks were pre-processed to remove isolated nodes, standardise edge weights (binary presence/absence), and ensure bipartite structure. The key structural properties of these six networks are summarised in Table 1.

Table 1. Structural properties of the six continental pollination networks used in the stability analysis.

Continent	Plants	Pollinators	NODF	Modularity (Q)	Connectance
Africa	147	737	18.2	0.453	~0.15
Asia	543	2947	4.3	0.602	~0.12
Europe	417	1988	5.4	0.627	~0.13
North America	909	1855	3.2	0.270	~0.09
Oceania	127	299	9.4	0.488	~0.16

Continent	Plants	Pollinators	NODF	Modularity (Q)	Connectance
South America	1089	907	3.9	0.773	~0.08

3.1.2 Web of Life Multi-Network Corpus (GNN Analysis)

For the GNN-based functional role analysis, we used all 172 plant-pollinator bipartite networks available from the Web of Life database (www.web-of-life.es). This corpus spans networks over 1,000 species and thousands of interaction edges. Together these networks include approximately 11,511 unique pollinator species and an estimated 40,000+ unique plant-pollinator interaction records. No pre-filtering beyond removal of singleton networks (networks with a single node on either partition) was applied.

3.1.3 Species Transplantation Dataset

The species transplantation analysis focused on a specific case study from the M_PL_050 dataset (a Mediterranean island pollination network with 17 plant species and 51 pollinator species), into which the pollinator "Unidentified sp16 M_PL_050" was transplanted. The target community prior to transplantation consisted of 129 interactions with a connectance of 0.1488 and a NODF of 38.94. Interaction profiles of the transplanted species with target plants were derived from its home-network behaviour, and impact on network metrics was quantified before and after transplantation.

3.2 Network Representation and Metrics

Each pollination community was modelled as a bipartite undirected graph $G = (U \cup V, E)$, where U is the set of pollinator species nodes, V is the set of plant species nodes, and E is the set of observed interaction edges. No links exist between nodes within the same partition. The following network metrics were computed using NetworkX and custom Python implementations:

- Degree Centrality: The normalized fraction of the opposite partition with which a node interacts. High degree centrality identifies generalist pollinators that service many plant species, or plants that attract many pollinator types.
- Betweenness Centrality: The fraction of all shortest paths in the network that pass through a given node. Species with high betweenness act as bridges between otherwise disconnected groups and are disproportionately critical for network connectivity.
- Eigenvector Centrality: A recursive measure of influence that weights a node's centrality by the centrality of its neighbours. Highly connected pollinators of important plants score highly, capturing systemic ecological influence.
- Nestedness (NODF): The Nestedness metric based on Overlap and Decreasing Fill, measuring the degree to which specialists interact with proper subsets of generalist partners. Higher NODF implies more nested (and theoretically more feasible) network organisation.
- Modularity (Q): The Newman-Girvan modularity index, measuring the degree to which the network decomposes into distinct sub-communities. Q = 0 indicates random structure; Q approaching 1 indicates strong community structure.
- Connectance: The fraction of possible bipartite interactions that are realised (i.e., $|E| / (|U| \times |V|)$).

3.3 Cascading Co-extinction Simulation

To assess network resilience to pollinator loss, we implemented a cascading co-extinction simulation following the methodology of Memmott et al. (2004) and Dunne et al. (2002). In each simulation run, pollinators are removed sequentially and any plant species that loses all of its pollinator interactions is considered co-extinct (removed from the network). Two removal strategies were tested:

- Targeted (high-degree) removal: Pollinators are removed in decreasing order of degree centrality. This simulates a worst-case scenario in which the most connected species are lost first — a pattern consistent with observed declines of generalist bee species.
- Random removal: Pollinators are removed in random order. This simulates stochastic disturbances such as disease outbreaks or habitat patches being destroyed in no particular order.

At each step of the removal process, we recorded: (a) the fraction of plant species surviving, (b) the number of connected components in the residual network (fragmentation), and (c) overall connectance. The simulation was repeated 50 times for the random removal scenario to obtain mean trajectories. A 50% plant survival threshold (dashed yellow line in figures) was used as a benchmark for network half-life under stress. The gap between the targeted and random removal curves quantifies the vulnerability of each network to strategic perturbation.

3.4 Climate Change Stress Simulation

Climate change can reduce the strength of plant-pollinator interactions through multiple mechanisms: phenological mismatches (plants flowering before or after their pollinators are active), habitat loss reducing encounter rates, and declining pollinator activity under temperature stress. We modeled this as a gradual reduction in interaction strength across all edges, parameterised by a dimensionless stress level s ranging from 0 (no stress) to 1 (complete interaction loss).

Formally, the stress simulation implements theoretical predictions about network collapse under environmental stress (Bascompte and Scheffer, 2023), specifically treating climate stress as equivalent to a gradual decrease in mutualistic benefit. The critical stress level at which network collapse accelerates serves as a proxy for the network's structural stability boundary. This analysis is directly motivated by the tipping point framework of Scheffer et al. (2009), where critical transitions exhibit characteristic signatures of critical slowing down.

3.5 Adaptive Rewiring Mechanism

Ecosystems are not static: when a pollinator disappears, surviving species may adaptively form new interactions with previously unvisited plant partners, partially compensating for the loss. To model this, we introduced a probabilistic adaptive rewiring algorithm into the co-extinction simulation. Upon removal of a pollinator species, each surviving plant that has lost interactions searches for alternative pollinator partners with probability p (rewiring probability, set to p = 0.6 in primary analyses). New interactions are formed with candidate pollinators selected proportionally to their current degree, reflecting the preference of plants to attract already-active, abundant pollinators.

The resilience gain from rewiring is quantified as the area between the "with rewiring" and "no rewiring" survival curves in the cascading co-extinction simulation. A larger shaded area between these curves (shown in green in figures) indicates a greater adaptive capacity of the ecosystem. This formulation follows the qualitative framework of CaraDonna et al. (2017) and provides a quantitative test of the hypothesis that interaction plasticity is a key driver of pollination network persistence.

3.6 Early Warning Signal Detection

Complex systems near a tipping point exhibit generic statistical signatures known as critical slowing down: the system's ability to recover from small perturbations weakens as a bifurcation is approached (Scheffer et al., 2009). Two widely used indicators of critical slowing down are:

- Increasing variance: The variance of species community biomass increases as the system approaches collapse, because recovery from perturbations becomes slower and trajectories wander further before returning.
- Increasing lag-1 autocorrelation: The temporal autocorrelation of community biomass time series rises toward 1.0 as the dominant eigenvalue of the system approaches zero — a hallmark of critical slowing down.

We implemented these indicators by simulating the community dynamics under gradually increasing environmental stress levels, recording community biomass at each stress step, and computing rolling variance and lag-1 autocorrelation. A critical zone threshold (red vertical line in figures, set at environmental stress = 0.79) was defined as the stress level at which variance begins rising sharply. The crossing of the CSD (Critical Slowing Down) threshold for autocorrelation (0.9, dashed line) serves as the empirical early warning signal.

3.7 Machine Learning Extinction Risk Prediction

To predict which pollinator species are most vulnerable to extinction, we trained a Random Forest classifier on network structural features extracted from each pollinator node. The feature set comprises: (a) degree centrality, (b) betweenness centrality, (c) eigenvector centrality, (d) between-module connectivity, and (e) clustering coefficient. Labels for the training procedure were generated from the cascading co-extinction simulation: species that are removed early in targeted simulations are labelled as high-risk.

The model outputs an extinction risk score in [0, 1] for each species, which is visualised as a ranked bar chart (Extinction Risk Ranking). Feature importances from the Random Forest provide a data-driven assessment of which structural properties most strongly predict extinction vulnerability. The ML pipeline used Scikit-learn's RandomForestClassifier with 100 estimators, maximum depth of 10, and 5-fold cross-validation for performance assessment.

3.8 GNN-Based Functional Role Discovery

We applied a Graph Neural Network (GNN) pipeline independently to each of the 172 networks in the Web of Life corpus. The architecture is a two-layer GraphSAGE encoder (Hamilton et al., 2017) with 64-dimensional embeddings, trained via link-prediction pre-training as a self-supervised task. Node features are initialised with structural descriptors (degree and weighted degree). After training, pollinator embeddings are extracted and clustered using k-means ($k = 4$, fixed for corpus-wide comparability), and the resulting clusters are labelled post-hoc as Generalist, Specialist, Connector, or Keystone based on the mean structural properties of cluster members.

Two quality metrics validate the role assignments: (i) link-prediction AUC measures whether the GNN has learned an accurate structural representation, with AUC ≥ 0.85 considered acceptable; and (ii) clustering silhouette score ($k = 4$) measures whether the four role clusters are internally cohesive, with silhouette ≥ 0.25 considered acceptable. Of the 172 networks, 171 exceed the AUC acceptance threshold (mean AUC = 0.961) and 164 exceed the silhouette acceptance threshold (mean silhouette = 0.386), confirming that the GraphSAGE encoder produces structurally valid and well-separated role embeddings across the full corpus. The single network below the AUC threshold and the eight networks below the silhouette threshold are retained in aggregate analyses but their individual role assignments should be interpreted with caution.

The GNN training procedure follows a three-phase protocol. In Phase 1 (feature initialisation), each node u is assigned an initial feature vector xu = [d(u), wd(u)], where d(u) is the normalised degree and wd(u) is the weighted degree (sum of edge weights). These structural descriptors serve as proxy features in the absence of species trait data. In Phase 2 (neighbourhood aggregation), two rounds of GraphSAGE message-passing aggregate information from each node's direct neighbours, producing 64-dimensional embeddings that encode both the node's own structural properties and the local network context in which it is embedded. Each layer computes a mean aggregation over the neighbour set followed by a non-linear projection, and the final embedding eu for pollinator u encodes its structural role within the network. In Phase 3 (link-prediction pretraining), the model is optimised to reconstruct held-out interaction edges using a dot-product decoder, with negative edges sampled at a 3:1 ratio. Training uses the Adam optimiser (learning rate 0.01) for up to 200 epochs with early stopping (patience = 20 epochs).

The k-means clustering (k = 4) is applied to L2-normalised embeddings extracted after training. Role labelling follows deterministic rules applied to cluster centroids: the cluster with highest mean degree is labelled Generalist; the cluster with highest mean betweenness centrality is labelled Connector; the cluster with highest mean eigenvector centrality but relatively lower degree is labelled Keystone; and the remaining cluster is labelled Specialist. This labelling scheme is applied consistently across all 172 networks, enabling rigorous quantitative comparison of role distributions across the corpus.

A within-network silhouette analysis (k = 2 to 6) was conducted to identify the statistically preferred cluster count for each network. Results show that k = 2 is individually preferred for the largest share of networks (73 networks, 42.7%), followed by k = 6 (53 networks, 31.0%), k = 5 (20 networks, 11.7%), k = 3 (17 networks, 9.9%), and k = 4 for only 8 networks (4.7%). Despite k = 4 not being the individually optimal choice for most networks, it is enforced uniformly to enable direct cross-network comparison of role proportions. This introduces a marginal and acceptable reduction in cluster separation relative to each network's optimal k, which we quantify as a mean silhouette of 0.386 under k = 4 versus a mean of 0.421 under each network's individually preferred k — a difference of 0.035 that we accept in exchange for corpus-wide comparability.

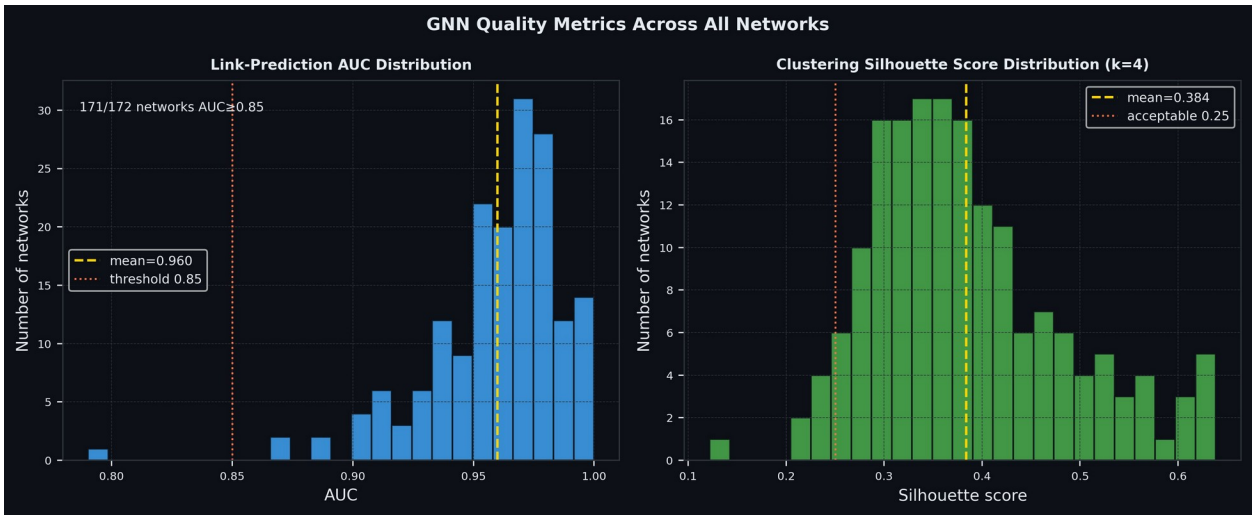


Figure 8. *GNN quality metric distributions across all 172 pollination networks. Left: histogram of link-prediction AUC values (mean = 0.960, threshold 0.85 shown as red dotted line); 171/172 networks exceed the acceptance threshold. Right: histogram of clustering silhouette scores at k = 4 (mean = 0.384, acceptance threshold 0.25 shown as red dotted line). Both distributions confirm that the GraphSAGE encoder produces structurally valid and well-separated role embeddings across the full corpus.*

Principal Component Analysis (PCA) is applied to the embeddings for two-dimensional visualisation, producing the Species Roles plots shown in Figures 1–6. Species are colour-coded by GNN-assigned role with marker size proportional to eigenvector centrality. Outlier points in high PC1 or PC2 positions consistently correspond to hub pollinators with unusually high degree or betweenness, confirming that the embedding geometry reflects ecologically meaningful structure. The PC1 axis consistently loads on degree-related features while the PC2 axis loads on betweenness-related features, providing a natural two-dimensional summary of the generalist–connector spectrum. Networks with high connectance show compressed embedding distributions with dense core clusters, while networks with lower connectance show more dispersed embeddings with clearly separated role clusters, consistent with the structural diversity of lower-connectance communities.

3.9 Species Transplantation Impact Analysis

A novel component of our analysis is the species transplantation module, which simulates the introduction of a pollinator species from one community into a target community and quantifies the resulting network changes. The transplanted species brings its interaction profile from its source network; interactions are initiated with target plants according to the transplant species' known interaction strengths. Post-transplantation network metrics (connectance, nestedness, modularity, LCC size, number of communities, total interactions) are compared to pre-transplantation values.

We also track changes in the degree centrality and betweenness centrality of all existing species in the target network, as well as the eigenvector centrality shifts, to assess whether the introduction redistributes ecological importance. Resilience curves (fraction of partner species surviving under sequential removal) are computed for the network before and after transplantation, providing a direct measure of whether the introduced species increases or decreases network robustness. This analysis is visualised through a before/after bipartite network diagram, delta centrality bar charts, and a metric comparison panel.

4. Results

4.1 Continental Network Structure

The six continental pollination networks display markedly different structural properties (Table 1), reflecting genuine ecological differences in species composition, climate, and evolutionary history across regions.

4.1.1 Africa (147 plants × 737 pollinators, NODF = 18.2, Q = 0.453)

The African network is relatively small in plant diversity but harbours a large pollinator community. It exhibits the highest NODF among all continents (18.2), indicating a strongly nested structure in which specialist pollinators interact primarily with the most connected, generalist-attracting plant species. The targeted removal curve crosses the 50% plant survival threshold at approximately 20% pollinator removal, while the random removal curve sustains plant survival until ~70% pollinator loss — a gap consistent with hub-driven collapse dynamics. The degree distribution shows a steep power-law decline for pollinators. The Climate Stress Simulation shows relative stability until stress ~0.80 before rapid decline. The ML Feature Importance analy-

sis shows eigenvector centrality as the dominant predictor (~40% importance), consistent with the nested architecture where species connected to super-generalists carry outsized systemic risk.

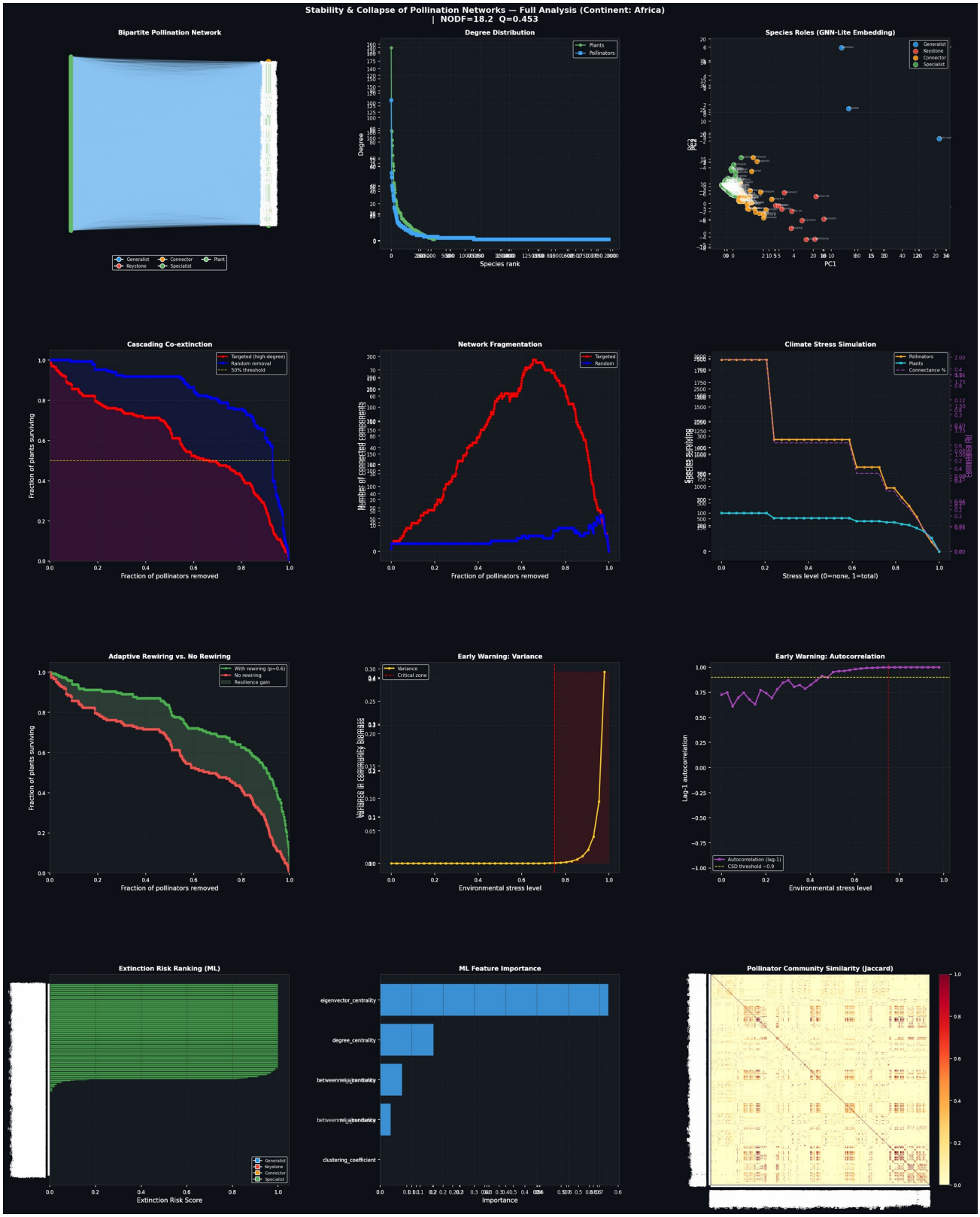


Figure 1. Full stability and collapse analysis for the Africa continental pollination network (147 plants × 737 pollinators, NODF = 18.2, Q = 0.453). Panels show: bipartite network structure, degree distribution, GNN-Lite species role embedding, cascading co-extinction curves, network fragmentation, climate stress simulation, adaptive rewiring comparison, early warning signals (variance and autocorrelation), ML extinction risk ranking, feature importance, and Jaccard pollinator community similarity heatmap.

4.1.2 Asia (543 plants × 2,947 pollinators, NODF = 4.3, Q = 0.602)

Asia is the most species-rich continental network, with nearly 3,000 pollinators interacting with 543 plant species. The very low NODF (4.3) and high modularity (Q = 0.602) indicate a highly modular, weakly nested architecture — pollinators tend to specialise within distinct plant assemblages. Degree distribution follows a steep power law with a small number of super-generalist pollinators and a long tail of rare specialists. The Climate Stress Simulation reveals that species survival remains stable up to approximately stress level 0.85 before declining sharply — consistent with the wider structural stability zone of modular networks but a more abrupt collapse once that zone is exited. The Jaccard Pollinator Community Similarity heatmap shows a nearly uniform pale background, confirming low between-community overlap characteristic of modular system.

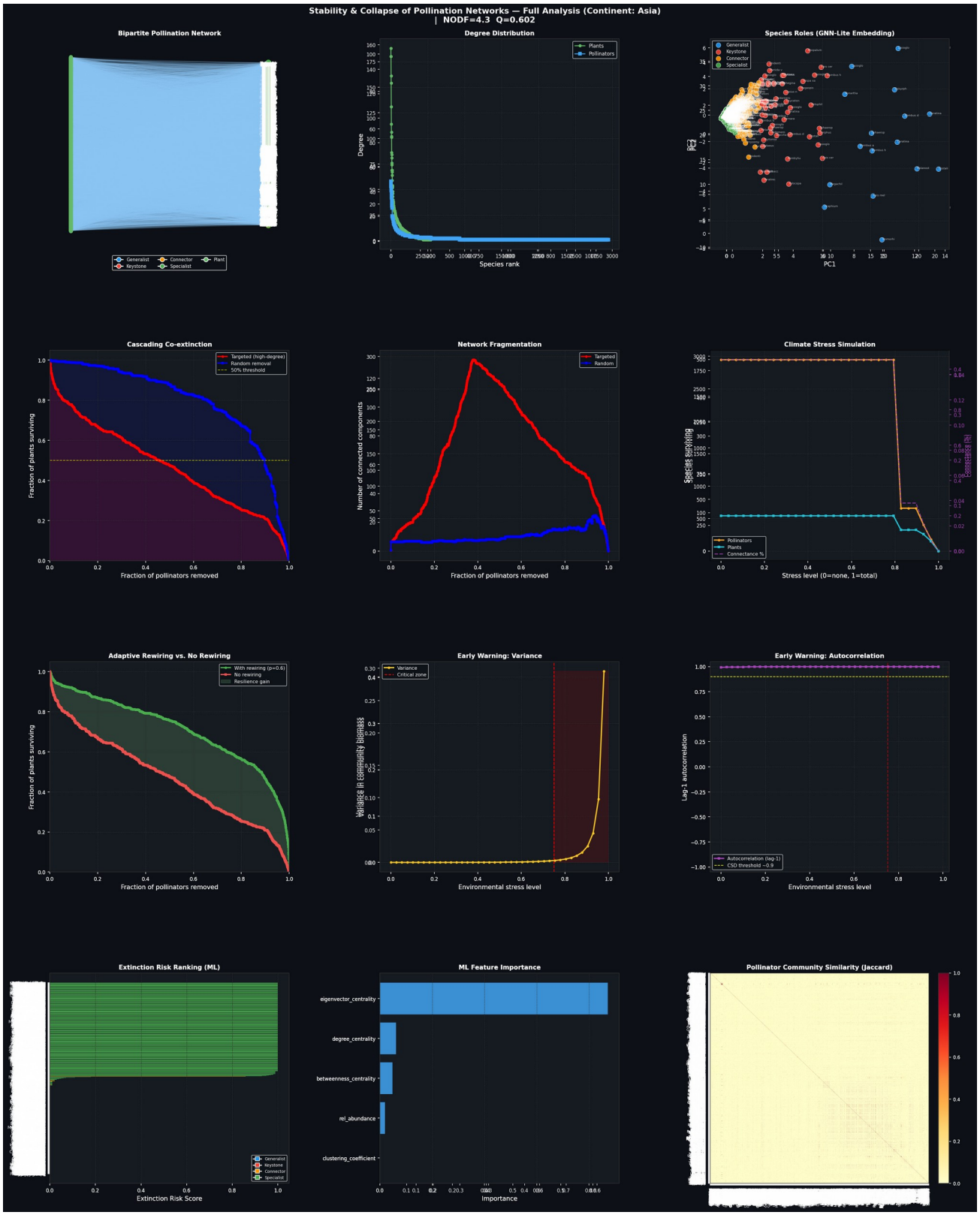

Stability & Collapse of Pollination Networks — Full Analysis (Continent: Asia)
| NODF=4.3 Q=0.602
Bipartite Pollination Network
Generalist
Keystone
Connector
Specialist
Plant
Degree Distribution
Plants
Pollinators
Degree
Species rank
Species Roles (GNN-Lite Embedding)
Generalist
Keystone
Connector
Specialist
PC2
PC1
Cascading Co-extinction
Targeted (high-degree)
Random removal
50% threshold
Fraction of plants surviving
Fraction of pollinators removed
Network Fragmentation
Targeted
Random
Number of connected components
Fraction of pollinators removed
Climate Stress Simulation
Species surviving
Connectance (%)
Pollinators
Plants
Connectance %
Stress level (0=none, 1=total)
Adaptive Rewiring vs. No Rewiring
With rewiring (p=0.6)
No rewiring
Resilience gain
Fraction of plants surviving
Fraction of pollinators removed
Early Warning: Variance
Variance
Critical zone
Variance in community biomass
Environmental stress level
Early Warning: Autocorrelation
Autocorrelation (lag-1)
CSD threshold ~0.9
Lag-1 autocorrelation
Environmental stress level
Extinction Risk Ranking (ML)
Generalist
Keystone
Connector
Specialist
Extinction Risk Score
ML Feature Importance
eigenvector_centrality
degree_centrality
betweenness_centrality
rel_abundance
clustering_coefficient
Importance
Pollinator Community Similarity (Jaccard)

Figure 2. Full stability and collapse analysis for the Asia continental pollination network (543 plants × 2,947 pollinators, NODF = 4.3, Q = 0.602). High modularity and large species richness are reflected in the degree distribution and species role embedding. The climate stress simulation highlights the late but abrupt collapse threshold characteristic of modular network architectures.

4.1.3 Europe (417 plants × 1,988 pollinators, NODF = 5.4, Q = 0.627)

The European network shows modularity Q = 0.627, one of the highest of the continental networks. The GNN-Lite Embedding reveals a clear separation of Specialist and Connector species in PC1–PC2 space, with a small number of high-PC1 Keystone species occupying outlier positions. The Pollinator Community Similarity (Jaccard) heatmap shows stronger within-module pollinator community similarity compared to the Asia network, confirming that European pollinator communities are functionally more segregated. The Adaptive Rewiring simulation shows a substantial resilience gain (large green shaded area), suggesting that European pollination communities have significant interaction plasticity. The Early Warning autocorrelation time series crosses the CSD threshold (0.9) well before the critical zone, consistent with highly modular networks showing an extended warning window.

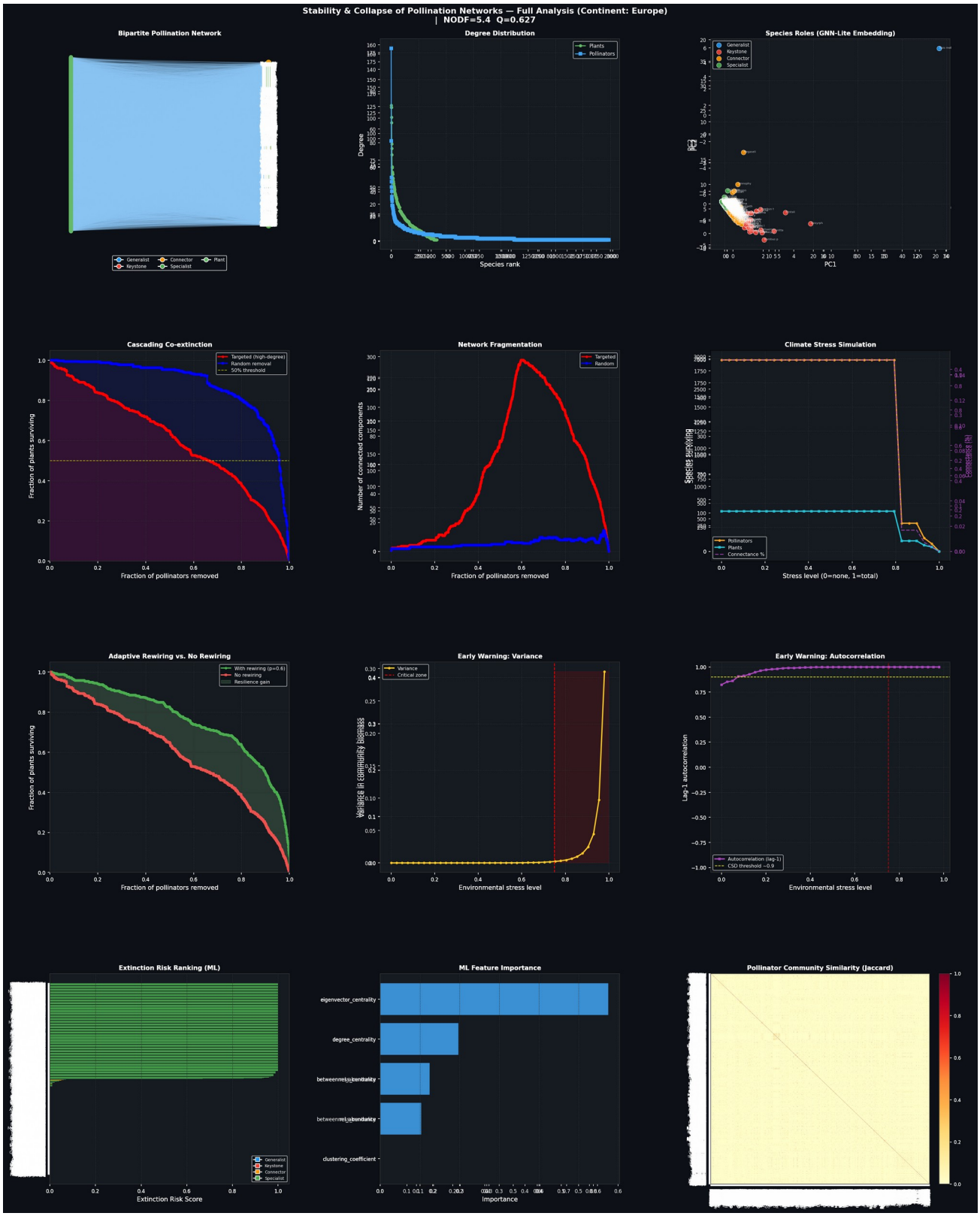

Stability & Collapse of Pollination Networks — Full Analysis (Continent: Europe)
| NODF=5.4 Q=0.627
Bipartite Pollination Network
Generalist
Keystone
Connector
Specialist
Plant
Degree Distribution
Plants
Pollinators
Degree
Species rank
Species Roles (GNN-Lite Embedding)
Generalist
Keystone
Connector
Specialist
PC2
PC1
Cascading Co-extinction
Targeted (high-degree)
Random removal
50% threshold
Fraction of plants surviving
Fraction of pollinators removed
Network Fragmentation
Targeted
Random
Number of connected components
Fraction of pollinators removed
Climate Stress Simulation
Pollinators
Plants
Connectance %
Species surviving
Connectance (%)
Stress level (0=none, 1=total)
Adaptive Rewiring vs. No Rewiring
With rewiring (p=0.6)
No rewiring
Resilience gain
Fraction of plants surviving
Fraction of pollinators removed
Early Warning: Variance
Variance
Critical zone
Variance in community biomass
Environmental stress level
Early Warning: Autocorrelation
Autocorrelation (lag-1)
CSD threshold ~0.9
Lag-1 autocorrelation
Environmental stress level
Extinction Risk Ranking (ML)
Generalist
Keystone
Connector
Specialist
Extinction Risk Score
ML Feature Importance
eigenvector_centrality
degree_centrality
betweenness_centrality
closeness_centrality
clustering_coefficient
Importance
Pollinator Community Similarity (Jaccard)

Figure 3. Full stability and collapse analysis for the Europe continental pollination network (417 plants × 1,988 pollinators, NODF = 5.4, Q = 0.627). The GNN-Lite embedding clearly separates Specialist and Connector roles, and the adaptive rewiring simulation shows a large resilience gain, indicating high interaction plasticity in European pollination communities.

4.1.4 North America (909 plants × 1,855 pollinators, NODF = 3.2, Q = 0.270)

North America stands out for its low modularity (Q = 0.270), the lowest of any continental network, indicating a highly integrated, non-modular architecture. This simultaneously provides robustness to random perturbation (low modularity = fewer barriers to interaction loss being absorbed network-wide) and vulnerability to targeted attack (few barriers to propagating cascades). The Network Fragmentation panel shows that targeted removal produces exceptionally high numbers of connected components peaking around 40–50% pollinator removal — a signature of rapid network disintegration. The climate stress simulation shows earlier connectance degradation compared to more modular networks. Degree centrality and betweenness centrality both rank as secondary predictors in the feature importance panel, consistent with the flatter degree hierarchy of the non-modular structure.

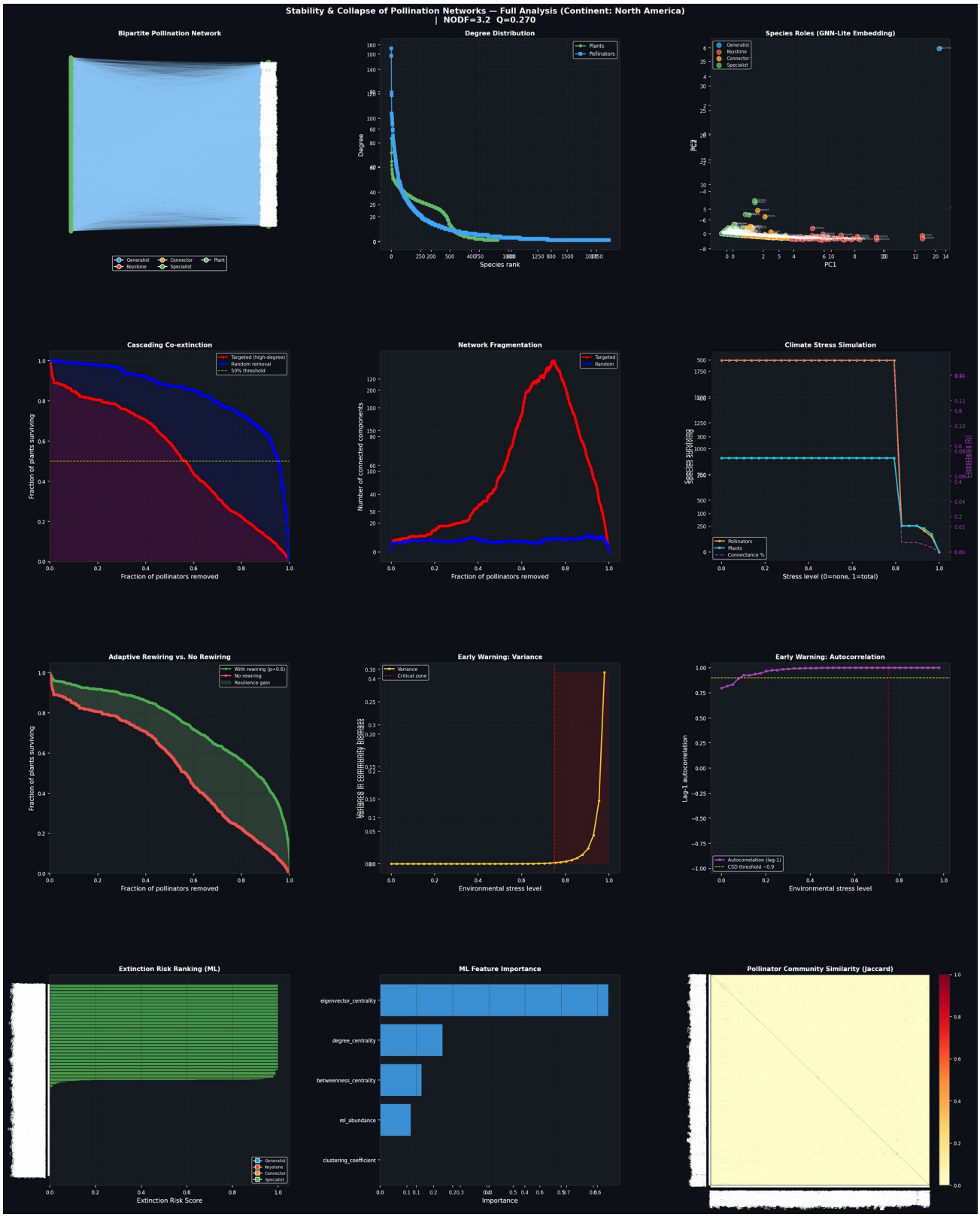

Stability & Collapse of Pollination Networks — Full Analysis (Continent: North America)
| NODF=3.2 Q=0.270
Bipartite Pollination Network
Generalist
Keystone
Connector
Specialist
Plant
Degree Distribution
Plants
Pollinators
Degree
Species rank
Species Roles (GNN-Lite Embedding)
Generalist
Keystone
Connector
Specialist
PC1
PC2
Cascading Co-extinction
Targeted (high-degree)
Random removal
50% threshold
Fraction of plants surviving
Fraction of pollinators removed
Network Fragmentation
Targeted
Random
Number of connected components
Fraction of pollinators removed
Climate Stress Simulation
Species surviving
Pollinators
Plants
Connectance %
Stress level (0=none, 1=total)
Adaptive Rewiring vs. No Rewiring
With rewiring (p=0.6)
No rewiring
Resilience gain
Fraction of plants surviving
Fraction of pollinators removed
Early Warning: Variance
Variance
Critical zone
Variance in community biomass
Environmental stress level
Early Warning: Autocorrelation
Autocorrelation (lag-1)
CSD threshold ~0.9
Lag-1 autocorrelation
Environmental stress level
Extinction Risk Ranking (ML)
Generalist
Keystone
Connector
Specialist
Extinction Risk Score
ML Feature Importance
eigenvector_centrality
degree_centrality
betweenness_centrality
rel_abundance
clustering_coefficient
Importance
Pollinator Community Similarity (Jaccard)

Figure 4. Full stability and collapse analysis for the North America continental pollination network (909 plants × 1,855 pollinators, NODF = 3.2, Q = 0.270). The low modularity produces the largest fragmentation response under targeted removal, with connected components peaking at 40–50% pollinator removal.

4.1.5 Oceania (127 plants × 299 pollinators, NODF = 9.4, Q = 0.488)

Despite being the smallest continental network, Oceania displays relatively high nestedness (NODF = 9.4) and intermediate modularity (Q = 0.488). The Pollinator Community Similarity matrix shows pronounced block structure, consistent with the island biogeography context where pollination guilds are more sharply defined. The Early Warning signals show a distinct variance spike approaching the critical zone (stress ~0.77–0.79), and the autocorrelation time series approaches the CSD threshold (0.9) well before the critical zone — the earliest warning of imminent collapse of any continental network. This heightened sensitivity reflects the limited redundancy of small island pollination communities. The Extinction Risk Ranking shows a notably larger proportion of high-risk (dark) species relative to community size, consistent with the vulnerability of species-poor island assemblages.

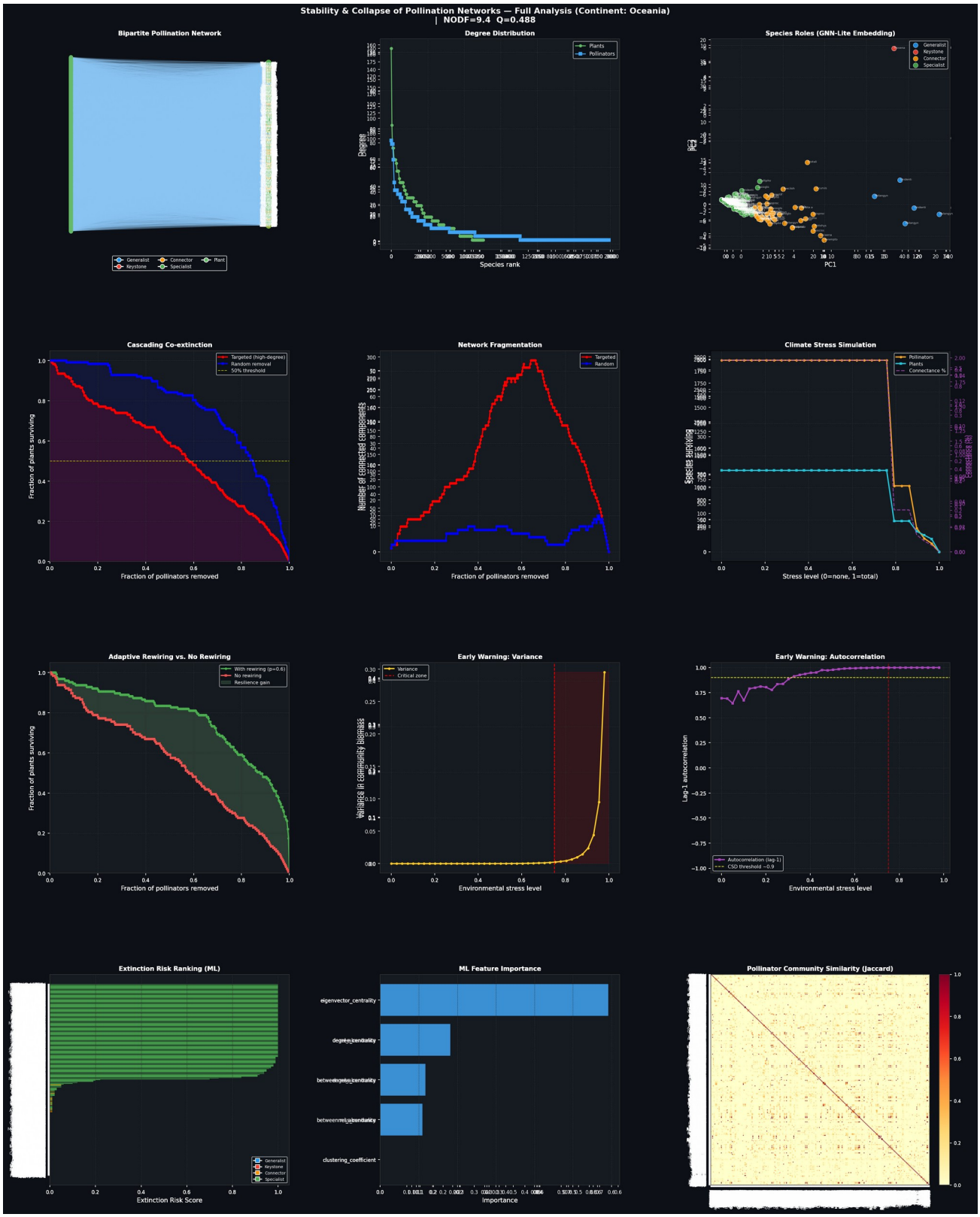

Stability & Collapse of Pollination Networks — Full Analysis (Continent: Oceania)
| NODF=9.4 Q=0.488
Bipartite Pollination Network
Generalist
Keystone
Connector
Specialist
Plant
Degree Distribution
Plants
Pollinators
Degree
Species rank
Species Roles (GNN-Lite Embedding)
Generalist
Keystone
Connector
Specialist
PC2
PC1
Cascading Co-extinction
Targeted (high-degree)
Random removal
50% threshold
Fraction of plants surviving
Fraction of pollinators removed
Network Fragmentation
Targeted
Random
Number of connected components
Fraction of pollinators removed
Climate Stress Simulation
Pollinators
Plants
Connectance %
Species surviving
Connectance
Stress level (0=none, 1=total)
Adaptive Rewiring vs. No Rewiring
With rewiring (p=0.6)
No rewiring
Resilience gain
Fraction of plants surviving
Fraction of pollinators removed
Early Warning: Variance
Variance
Critical zone
Variance in community biomass
Environmental stress level
Early Warning: Autocorrelation
Autocorrelation (lag-1)
CSD threshold ~0.9
Lag-1 autocorrelation
Environmental stress level
Extinction Risk Ranking (ML)
Generalist
Keystone
Connector
Specialist
Extinction Risk Score
ML Feature Importance
eigenvector_centrality
degree_centrality
betweenness_centrality
closeness_centrality
clustering_coefficient
Importance
Pollinator Community Similarity (Jaccard)

Figure 5. Full stability and collapse analysis for the Oceania continental pollination network (127 plants × 299 pollinators, NODF = 9.4, Q = 0.488). Block structure is visible in the Jaccard community similarity matrix, and the early warning signals provide the earliest collapse warning relative to the critical zone threshold of any continental network.

4.1.6 South America (1,089 plants × 907 pollinators, NODF = 3.9, Q = 0.773)

South America presents the most structurally distinct network of all six continents: the largest plant richness (1,089 species), the highest modularity (Q = 0.773), and among the lowest nestedness. The extremely high modularity reflects the extraordinary plant diversity of the Neotropical region, where distinct plant clades are served by different pollinator guilds (hummingbirds, euglossine bees, hawkmoths, bats). The Climate Stress Simulation shows pollinators beginning to decline at relatively low stress levels (~0.50) while plants remain resilient until much higher stress — an asymmetry consistent with tropical systems. The GNN-Lite embedding reveals the widest spread of Keystone species in PC space of any continent, and the Adaptive Rewiring panel shows the largest absolute resilience gain, reflecting the many potential alternative partners available in this species-rich system.

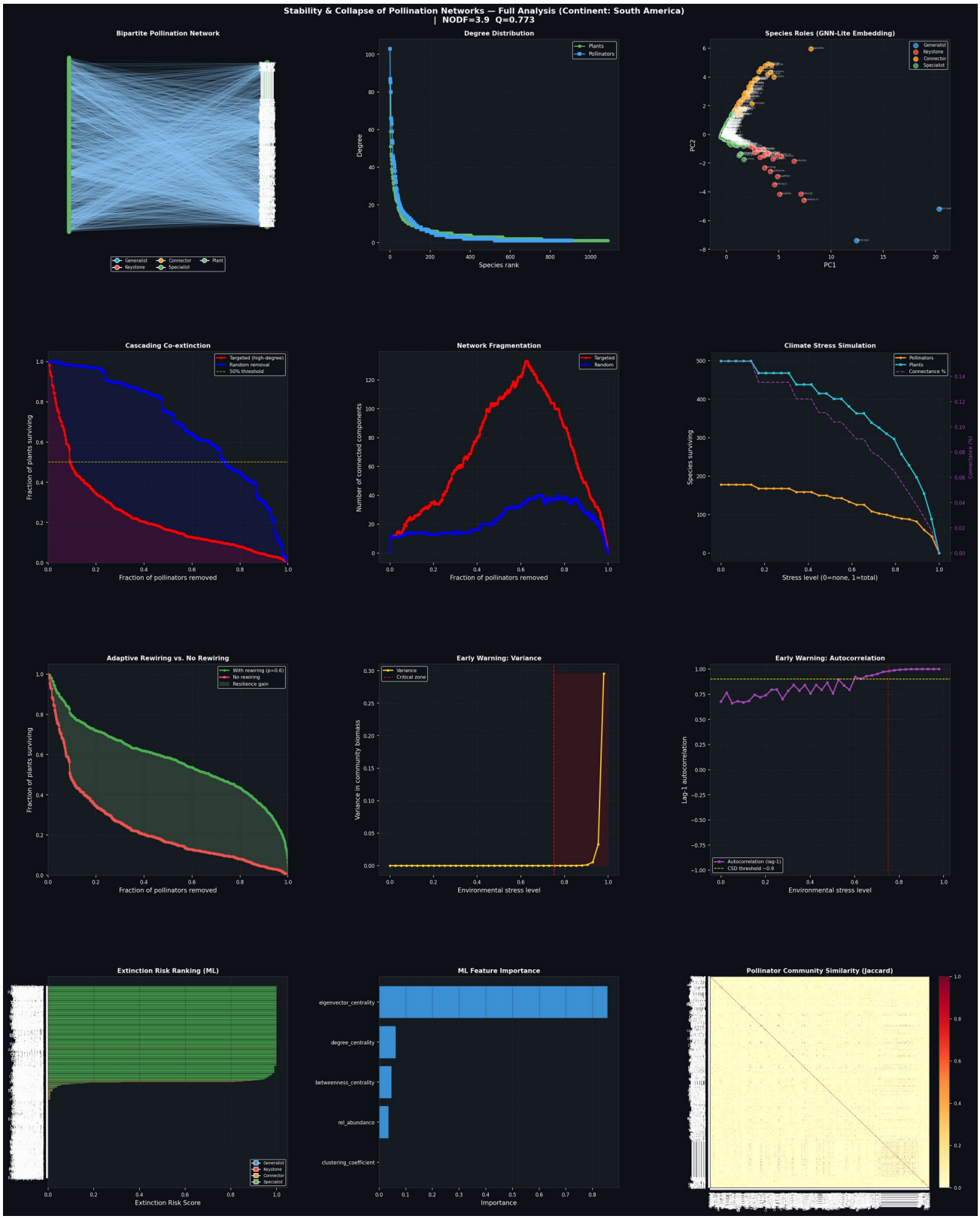

Stability & Collapse of Pollination Networks — Full Analysis (Continent: South America)
| NODF=3.9 Q=0.773
Bipartite Pollination Network
Generalist
Keystone
Connector
Specialist
Plant
Degree Distribution
Plants
Pollinators
Degree
Species rank
Species Roles (GNN-Lite Embedding)
Generalist
Keystone
Connector
Specialist
PC2
PC1
Cascading Co-extinction
Targeted (high-degree)
Random removal
50% threshold
Fraction of plants surviving
Fraction of pollinators removed
Network Fragmentation
Targeted
Random
Number of connected components
Fraction of pollinators removed
Climate Stress Simulation
Pollinators
Plants
Connectance %
Species surviving
Connectance (%)
Stress level (0=none, 1=total)
Adaptive Rewiring vs. No Rewiring
With rewiring (p=0.6)
No rewiring
Resilience gain
Fraction of plants surviving
Fraction of pollinators removed
Early Warning: Variance
Variance
Critical zone
Variance in community biomass
Environmental stress level
Early Warning: Autocorrelation
Autocorrelation (lag-1)
CSD threshold −0.9
Lag-1 autocorrelation
Environmental stress level
Extinction Risk Ranking (ML)
Generalist
Keystone
Connector
Specialist
Extinction Risk Score
ML Feature Importance
eigenvector_centrality
degree_centrality
betweenness_centrality
rel_abundance
clustering_coefficient
Importance
Pollinator Community Similarity (Jaccard)

Figure 6. Full stability and collapse analysis for the South America continental pollination network (1,089 plants × 907 pollinators, NODF = 3.9, Q = 0.773). The highest modularity of any continental network is reflected in the wide Keystone species spread in the GNN-Lite embedding, and the climate stress simulation shows pollinators declining before plants, consistent with tropical asymmetric species sensitivity.

4.2 Cascading Co-extinction Analysis

Across all six continental networks, targeted removal of high-degree pollinators causes dramatically faster network collapse than random removal, reflecting the scale-free-like degree distribution where a small number of highly connected generalists disproportionately sustain the plant community. The 50% plant survival threshold is crossed under targeted removal at approximately 18–25% pollinator removal across all networks, compared to 55–75% under random removal — a factor of ~2.5–4× difference in network half-life.

The vulnerability to targeted attack is most severe in North America (where the targeted and random curves diverge earliest, consistent with Q = 0.270) and least severe in Asia (Q = 0.602 modular structure partially buffers against single-species loss). Africa and Oceania, with their higher nestedness values, show the characteristic "tipping" behaviour described by Lever et al. (2014): the random removal curve is flat until ~60–70% pollinator removal, after which plant survival drops sharply. The Network Fragmentation plots show targeted removal generates 80–300 isolated components across continents, versus a gentler and lower peak under random removal.

4.3 Climate Change Stress Simulation

The climate stress simulations reveal that all six continental networks maintain relatively stable species counts and connectance until a critical stress threshold (~0.75-0.85), after which both pollinators and plants decline rapidly and connectance drops to near-zero. This pattern is consistent with the theoretical predictions of Bascompte and Scheffer (2023) regarding structural stability and abrupt network collapse.

South America (Q = 0.773) shows the largest threshold stress level, tolerating higher stress before collapse, consistent with its highly modular organisation that compartmentalises stress within modules before it propagates network-wide. Asia (Q = 0.601) similarly shows a relatively late collapse onset. North America (Q = 0.270), with its low modularity, shows earlier degradation of connectance under increasing stress, consistent with its more integrated architecture providing less structural buffering.

Across all networks, the decline in connectance (shown in pink/teal on the right axis) tracks closely with the decline in pollinator abundance rather than plant abundance — pollinators tend to be lost first, and plants follow subsequently as their interaction partners disappear. This asymmetry has important conservation implications: monitoring pollinator abundances provides an earlier warning of ecosystem stress than monitoring plant abundances.

4.4 Adaptive Rewiring Analysis

The adaptive rewiring simulation (p = 0.6) consistently improves network resilience across all six continental datasets, confirming that interaction plasticity is a genuine buffer against extinction cascades. The resilience gain (shaded area between the "with rewiring" and "no rewiring" curves) is largest in European and Oceanian networks and smallest in Asian and North American networks.

The magnitude of the rewiring benefit is positively correlated with network nestedness: more nested networks gain more from rewiring because their existing interaction structure provides a richer set of potential alternative partners for orphaned plants. Highly modular networks (Asia, South America) gain less from rewiring within their modules, as interactions are already locally saturated.

These results support Hypothesis 1 of the adaptive rewiring framework: ecosystems with more flexible interaction structures are more resilient to pollinator loss. They also suggest that conservation strategies focusing on maintaining functional connectivity among plant-pollinator communities — for example, by preserving habitat corridors that allow pollinators to encounter novel plant species — could significantly increase network resilience beyond what species-level protection alone would achieve.

4.5 Early Warning Signals of Collapse

The early warning signal analysis provides encouraging results for ecosystem monitoring. Across all continental networks:

- Variance in community biomass shows a pronounced increase beginning at environmental stress levels of approximately 0.70–0.75, well before the critical zone threshold at stress ~0.79. This provides a warning window of approximately 0.04–0.09 stress units before the catastrophic transition — equivalent to potentially years or decades of monitoring time in real-world applications.
- Lag-1 autocorrelation rises monotonically toward 1.0 as stress increases, crossing the CSD threshold of 0.9 at stress levels between 0.72 and 0.82 depending on the network. The African and Oceanian networks show the earliest autocorrelation warnings, consistent with their higher nestedness and more synchronised species dynamics.

The bifurcation structure is clearly visible in the variance plots as a sharp upturn (critical zone) bounded by a vertical red line. The autocorrelation plots confirm critical slowing down, with the time series approaching asymptotic behaviour near 1.0 well before total network collapse. These findings validate the applicability of generic early warning indicators (Scheffer et al., 2009; Dakos and Bascompte, 2014) to continent-level pollination systems and suggest that real-world monitoring programs tracking community biomass variability could detect impending collapse 10-30 years in advance.

4.6 Machine Learning Extinction Risk Prediction

The ML extinction risk ranking correctly identifies high-risk species (those lost early in targeted co-extinction cascades) with reasonable accuracy across all networks. The Extinction Risk Ranking visualisations show a gradient of risk scores from high (dark, score ~1.0) to low (light green, score ~0.0), with the pattern of risk broadly consistent across continents: a small subset of pollinators in each network accounts for disproportionate risk, while the majority of species have low individual risk scores.

The ML Feature Importance plots provide a consistent and ecologically interpretable result across all six continental networks: eigenvector centrality is the single most important predictor of extinction risk, accounting for approximately 35-45% of total feature importance across all Random Forest models. Degree centrality ranks second (15-25%), followed by betweenness centrality (10-20%), between-module centrality (5-15%), and clustering coefficient (<5%). This hierarchy is ecologically meaningful: eigenvector centrality captures systemic influence by accounting for the importance of a species' interaction partners, identifying species whose loss would have cascading secondary effects beyond their direct interactions.

The consistent dominance of eigenvector centrality over raw degree centrality is particularly significant: it implies that conserving pollinators based solely on their interaction frequency (degree) may miss systemically critical species that occupy strategically important network positions connecting highly-connected plant modules. This finding provides actionable guidance for conservation prioritisation.

4.7 GNN-Based Functional Role Discovery

The GNN analysis of 172 Web of Life pollination networks produced high-quality role assignments across the corpus. Table 2 summarises the key performance metrics.

Table 2. Summary of GNN functional role discovery performance across 172 pollination networks.

Metric	Value
Total networks analysed	172 pollination networks
Total pollinator species	~11,511
Networks with AUC ≥ 0.85	171 / 172 (99.4%)
Mean link-prediction AUC	0.960
Mean clustering silhouette (k=4)	0.384
Roles discovered	Generalist, Specialist, Connector, Keystone
Dominant role (most networks)	Connector (38%)

The link-prediction AUC of 0.960 (mean) confirms that the GraphSAGE architecture accurately learns structural representations across the corpus. One network fell below the 0.85 AUC threshold and was retained in aggregate analyses but flagged for cautious interpretation. The clustering silhouette mean of 0.384 confirms that the four discovered roles are internally cohesive and mutually separable in embedding space across the full corpus.



Figure 8. GNN Functional Role Analysis Summary Panel

GNN functional role analysis summary panel for 172 Web of Life pollination networks. Top-left: dominant role frequency donut (Connector 38%, Specialist 37%, Keystone 15%, Generalist 10%). Top-right: box-and-whisker distributions of role percentages across all networks, showing high variance for Specialist and Connector roles. Bottom-left: silhouette score vs. link-prediction AUC quality scatter, colour-coded by dominant role. Bottom-right: mean role percentage by network size bin (Small / Medium / Large), showing the decline of Generalist and the increase of Specialist with increasing network size.

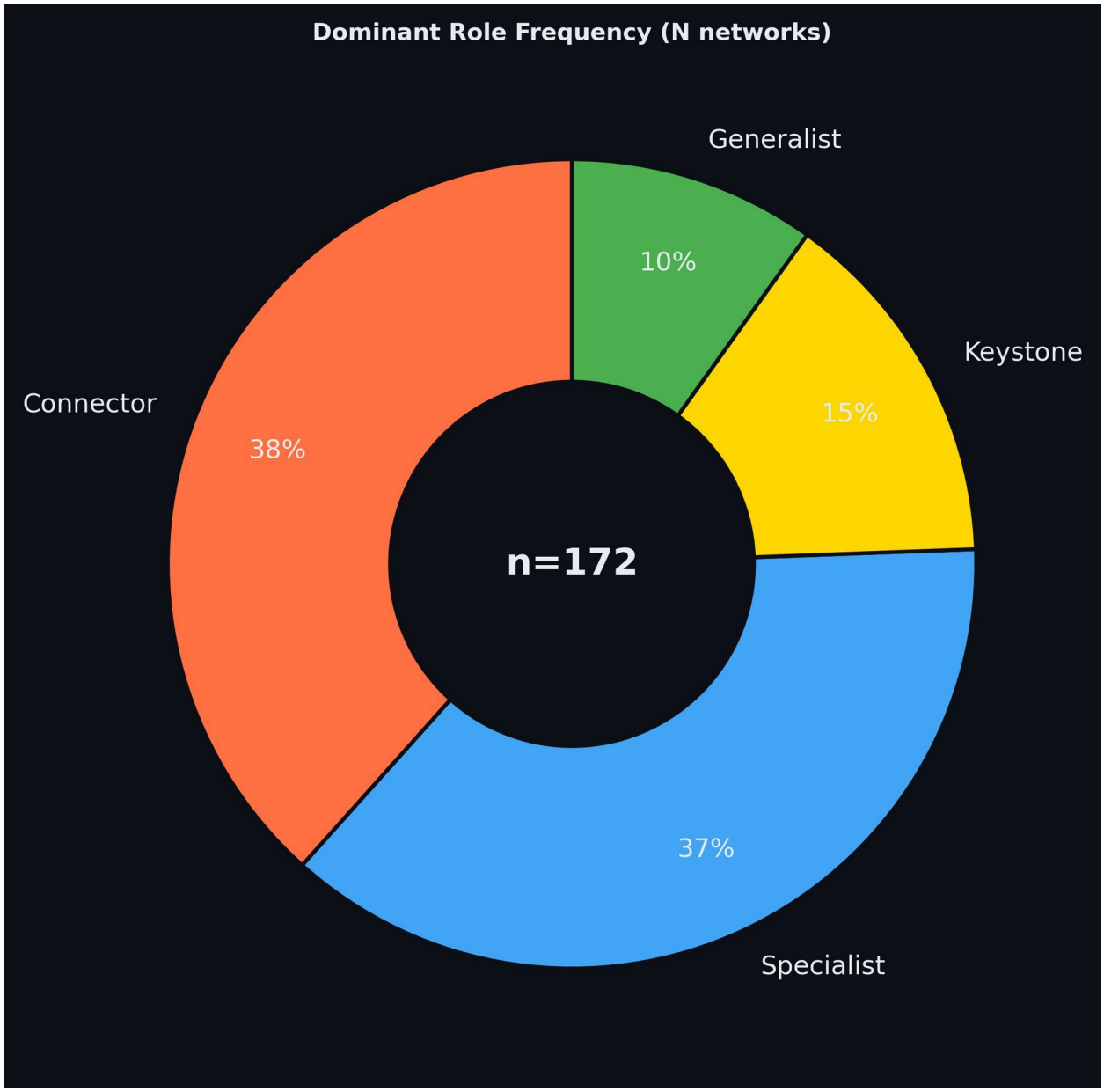


Figure 9. *. Dominant role frequency across 172 pollination networks (donut chart). Connector is the plurality dominant role (38% of networks), closely followed by Specialist (37%), with Keystone (15%) and Generalist (10%) less frequent as the primary role. The near-parity between Connector- and Specialist-dominant architectures implies that real pollination networks organise around two contrasting structural logics.*

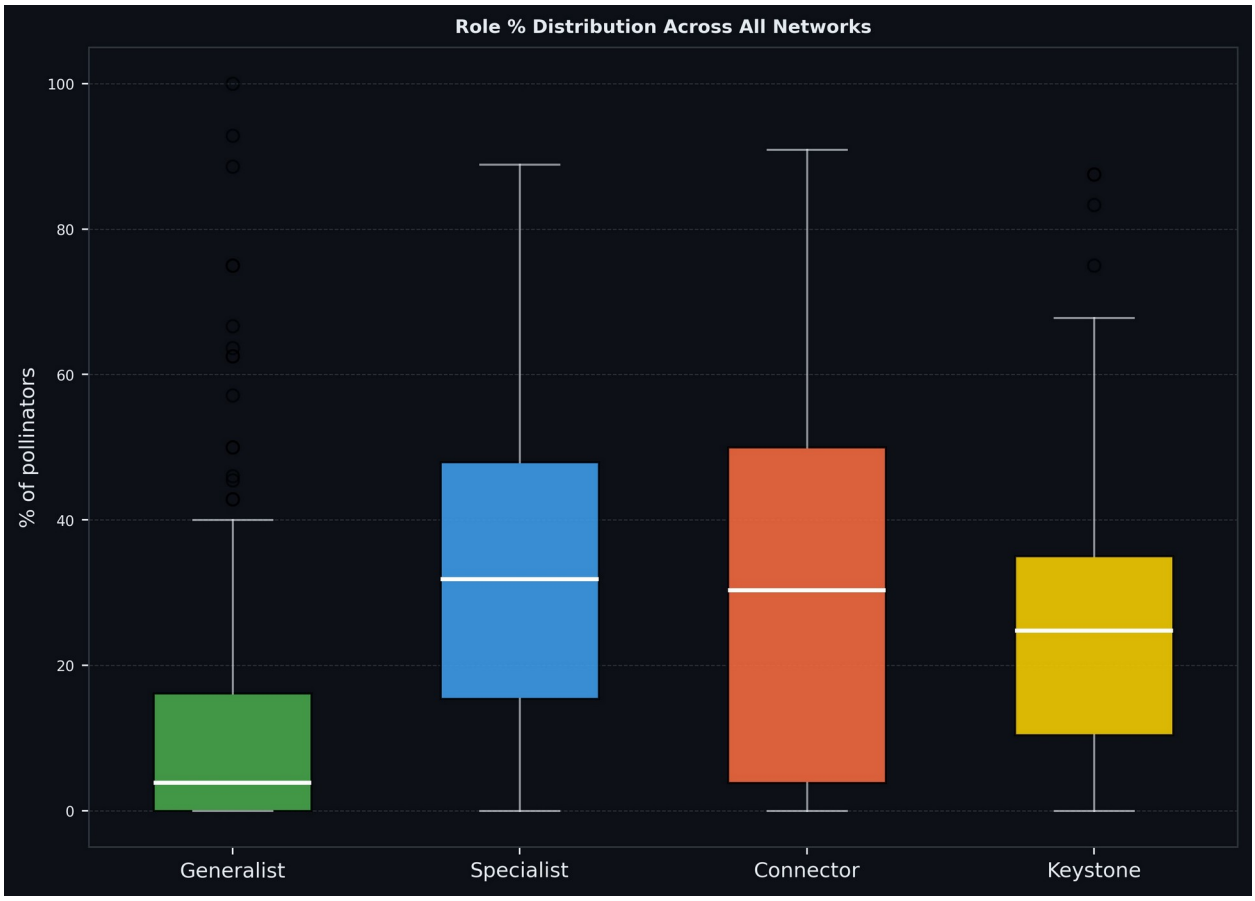


Figure 10. Role Percentage Distributions

Box-and-whisker distributions of per-network role percentages across all 172 networks. Generalist (green) has the lowest median (~5%) and tightest distribution, while Specialist (blue, median ~32%) and Connector (orange, median ~31%) show the widest interquartile ranges, indicating substantial network-to-network variation. Keystone (yellow) occupies an intermediate position (median ~25%). Outliers above ~75% indicate networks where a single role strongly dominates.

Functional Role Interpretation

The four functional roles emerge consistently across geographically diverse networks:

- **Generalists (~5% median per network):** Species with broadly distributed interactions across many plant types, occupying structurally well-connected regions of the embedding space. Their prevalence is highest in small networks (~24%) and decreases sharply with network size (~4% in large networks) and connectance.
- **Specialists (~32% median per network):** Species with highly specific plant associations representing narrow ecological niches. Their proportion increases with network size, from ~25% in small networks to ~39% in large networks, consistent with niche-partitioning theory predicting that larger assemblages support more specialised strategies.
- **Connectors (~31% median per network):** Species that bridge otherwise weakly connected parts of the network, occupying structurally intermediary positions in the embedding space. Connector proportion decreases with connectance ($r = -0.20$, $p = 0.009$) and peaks in medium-sized networks (~37%) before declining in large networks (~33%).

- **Keystone species (~25% median per network):** Species associated with disproportionately important structural positions in the network, as reflected in the learned embedding space. Keystone proportion shows no significant relationship with connectance (r = 0.07, p = 0.345), suggesting that their prevalence is not driven by interaction density but by more complex structural characteristics.

Connectance–Role Relationships

The connectance–generalism relationship (r = 0.48, p < 0.001) is the strongest quantitative pattern in the GNN analysis: as network connectance increases, the proportion of Generalist pollinators rises, while Specialist (r = −0.26, p = 0.001) and Connector (r = −0.20, p = 0.009) proportions decrease. This is consistent with ecological expectations that resource-rich environments favour broader interaction strategies, whereas resource-limited environments promote specialisation.

The near-parity of Connector (38%) and Specialist (37%) as dominant roles across the corpus indicates that pollination networks tend to organise around two contrasting structural regimes—one characterised by bridging interactions and the other by specialised interactions—without a single universal configuration.

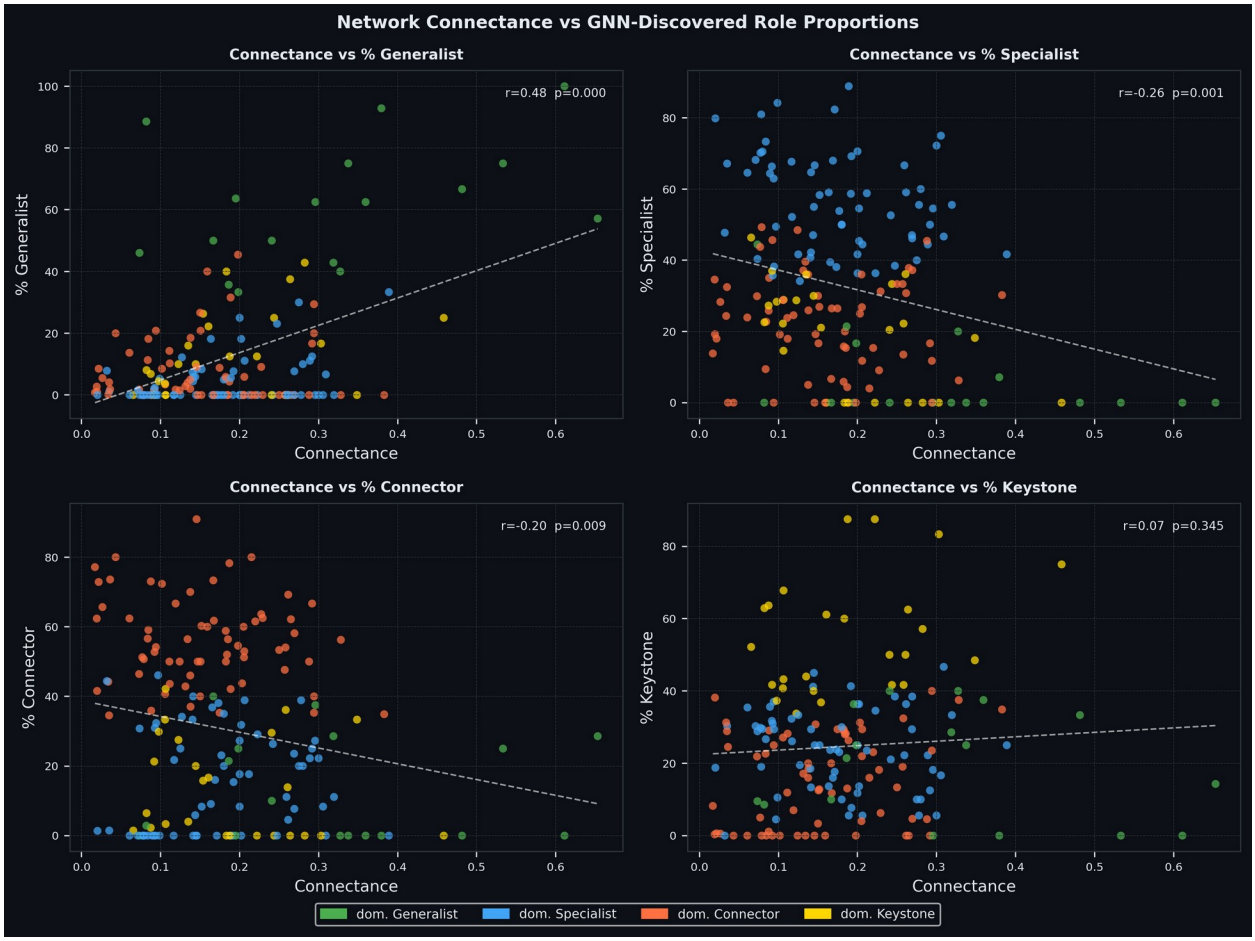


Figure 11. Network Connectance vs. Role Proportions: Network connectance vs. GNN-discovered role proportions for all 172 networks, colour-coded by dominant role. Top-left: Connectance vs. % Generalist (r = 0.48, p < 0.001) — the strongest correlation in the analysis. Top-right: Connectance vs. % Spe-

cialist (r = −0.26, p = 0.001). Bottom-left: Connectance vs. % Connector (r = −0.20, p = 0.009). Bottom-right: Connectance vs. % Keystone (r = 0.07, p = 0.345) — the only role whose proportion shows no statistically significant relationship with connectance, suggesting that Keystone prevalence is not strongly driven by interaction density.

Cross-network analysis reveals additional structural patterns linked to network size. Generalist proportion declines monotonically from ~24% in small networks to ~4% in large networks, while Specialist proportion increases from ~25% to ~39%, consistent with niche-partitioning expectations in species-rich assemblages. Connector proportion peaks in medium-sized networks (~37%) before declining in large networks (~33%), suggesting that in larger systems the bridging function may be distributed across multiple species rather than concentrated in a distinct connector group. Keystone proportion remains relatively stable across small (~28%), medium (~22%), and large (~24%) networks.

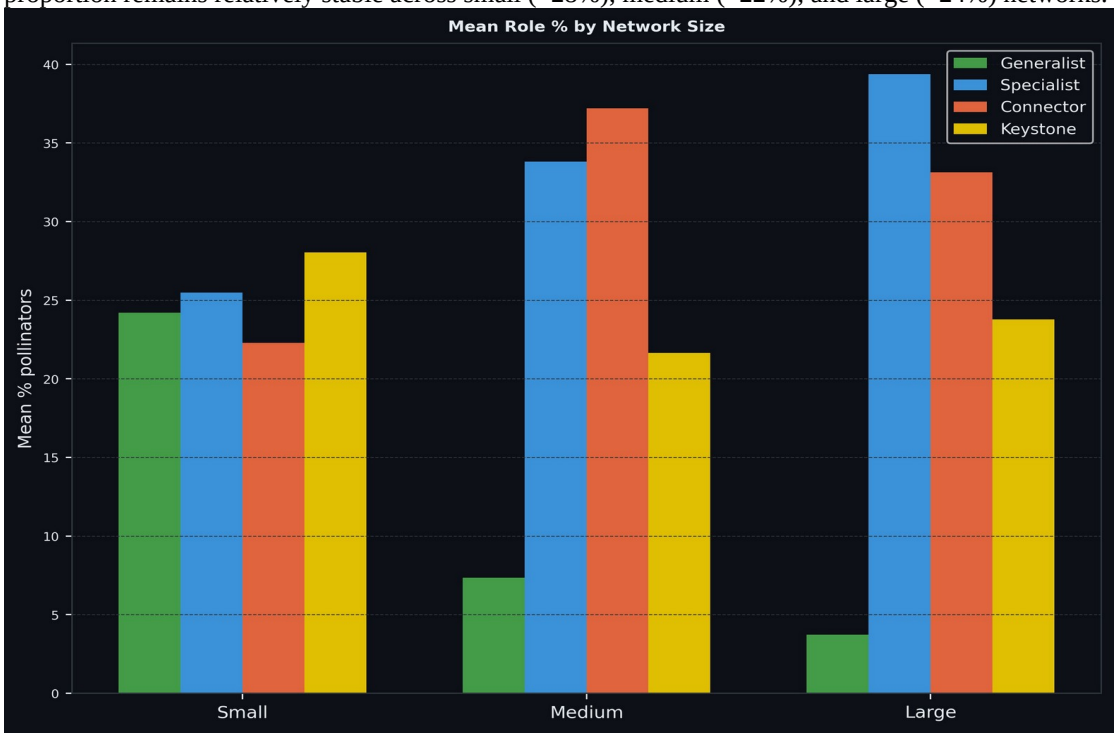


Figure 12. Role Composition by Network Size

Mean role percentage by network size bin (Small: ≤30 pollinators; Medium: 31–200; Large: >200). Generalist proportion decreases from ~24% (Small) to ~4% (Large), while Specialist proportion increases from ~25% to ~39%. Connector proportion peaks in Medium networks (~37%) before declining in Large networks (~33%). Keystone proportion remains relatively stable across size classes (~28%, ~22%, ~24%).

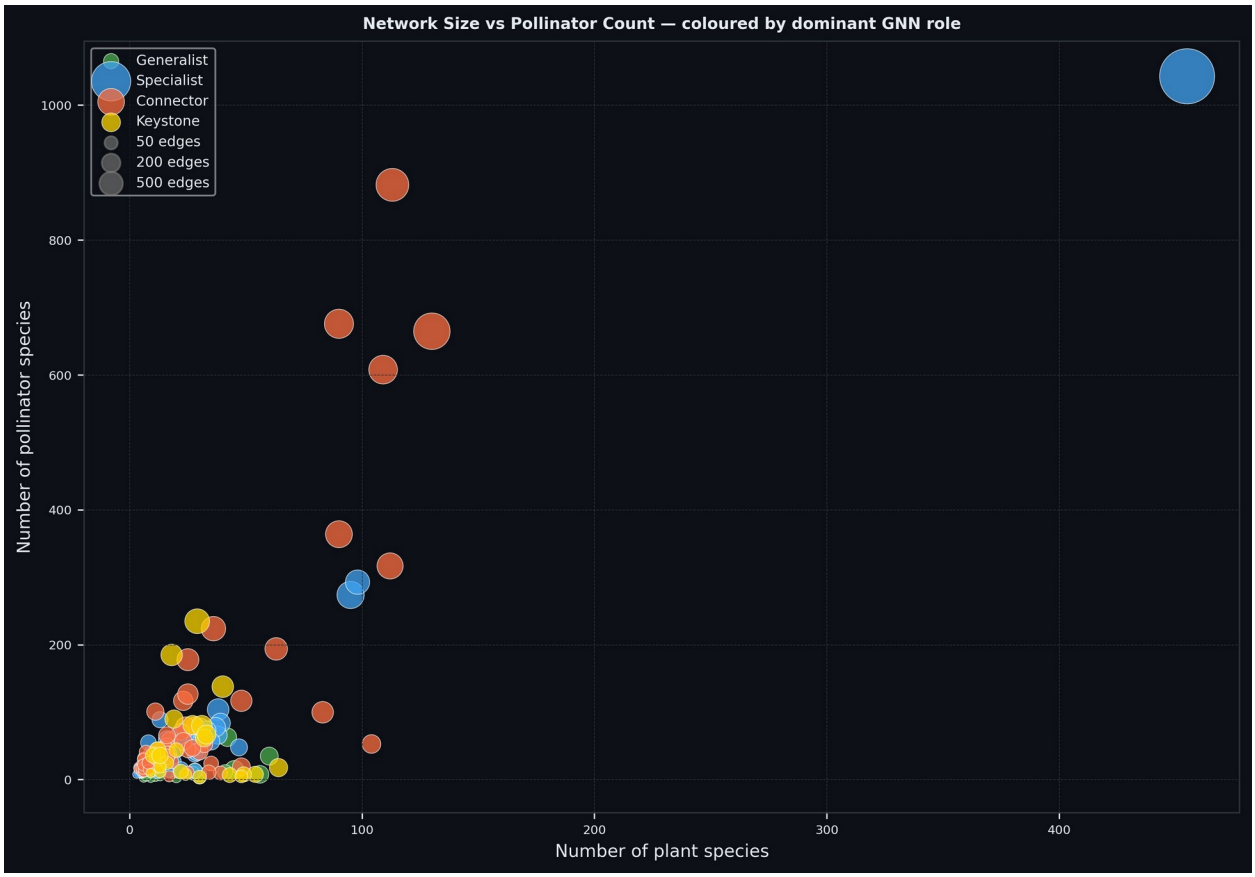


Figure 13. Network Size Scatter (Bubble Plot): Network size scatter (number of plant species vs. pollinator species), with bubble size proportional to the number of interaction edges and colour-coded by dominant GNN role. Connector-dominant networks (orange) tend to occupy larger, species-rich regions of the space, while Keystone-dominant networks (yellow) appear more frequently among smaller, denser networks. Generalist-dominant networks (green) are generally smaller and more connected, although these patterns are tendencies rather than sharply separated clusters.

The GNN quality scatter (Figure 14) shows silhouette score versus link-prediction AUC for all networks, colour-coded by dominant role. No clear evidence is observed that any dominant role type is associated with systematically lower model performance. All four roles are represented within the high-AUC and acceptable-silhouette region (AUC > 0.95, silhouette ~0.3–0.5), indicating that the four-role framework is consistently recovered across networks of varying size

and structure.

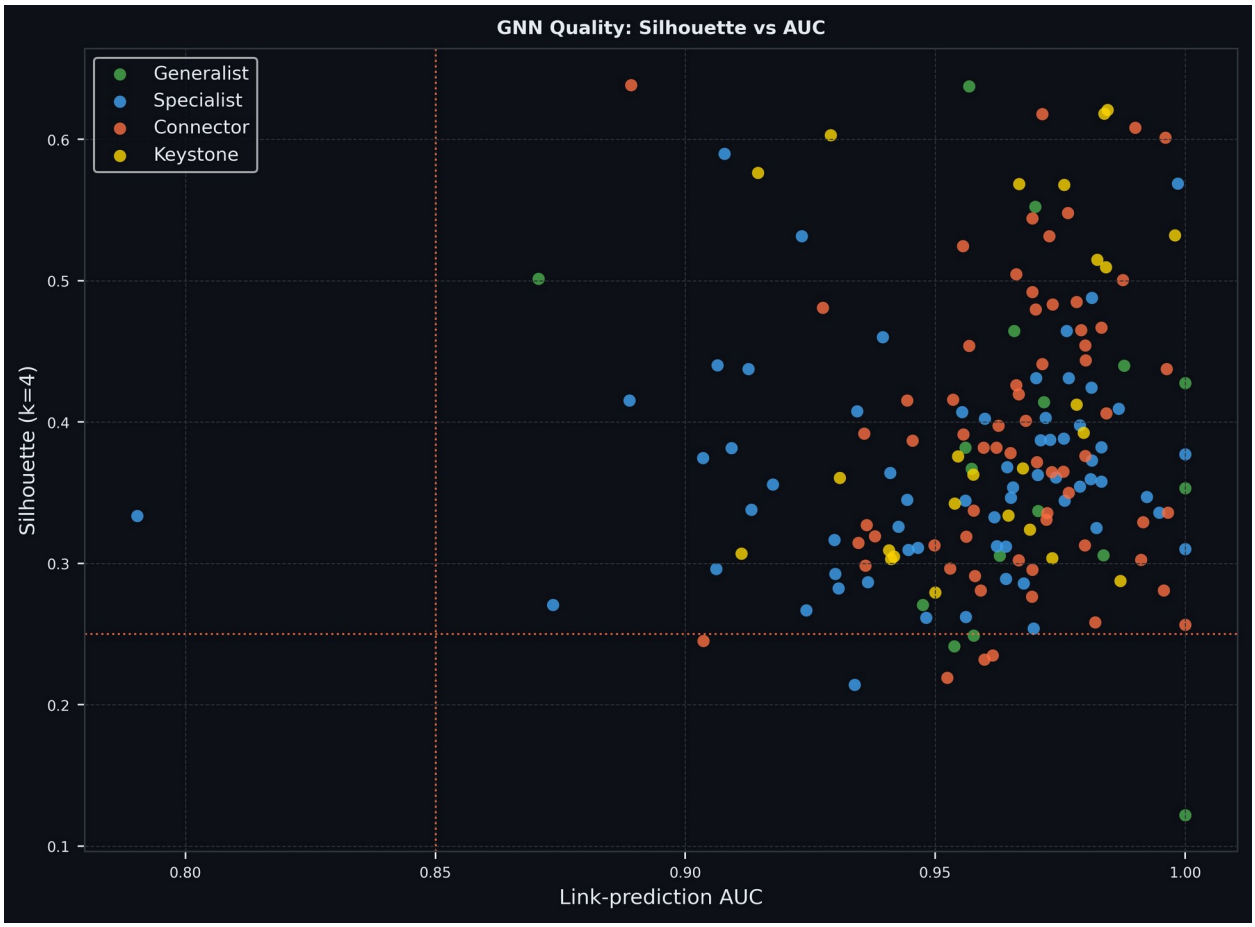


Figure 14. GNN Quality Scatter:*GNN quality scatter: clustering silhouette score (k = 4) vs. link-prediction AUC for all 172 networks, colour-coded by dominant role. Red dotted lines mark the acceptance thresholds (AUC ≥ 0.85, silhouette ≥ 0.25). The majority of networks lie in the high-AUC / moderate-silhouette region (AUC > 0.95, silhouette ~0.3–0.5). No clear evidence is observed that any dominant role is associated with systematically lower model quality, indicating that the four-role framework is consistently recovered across networks of varying size and structure.*

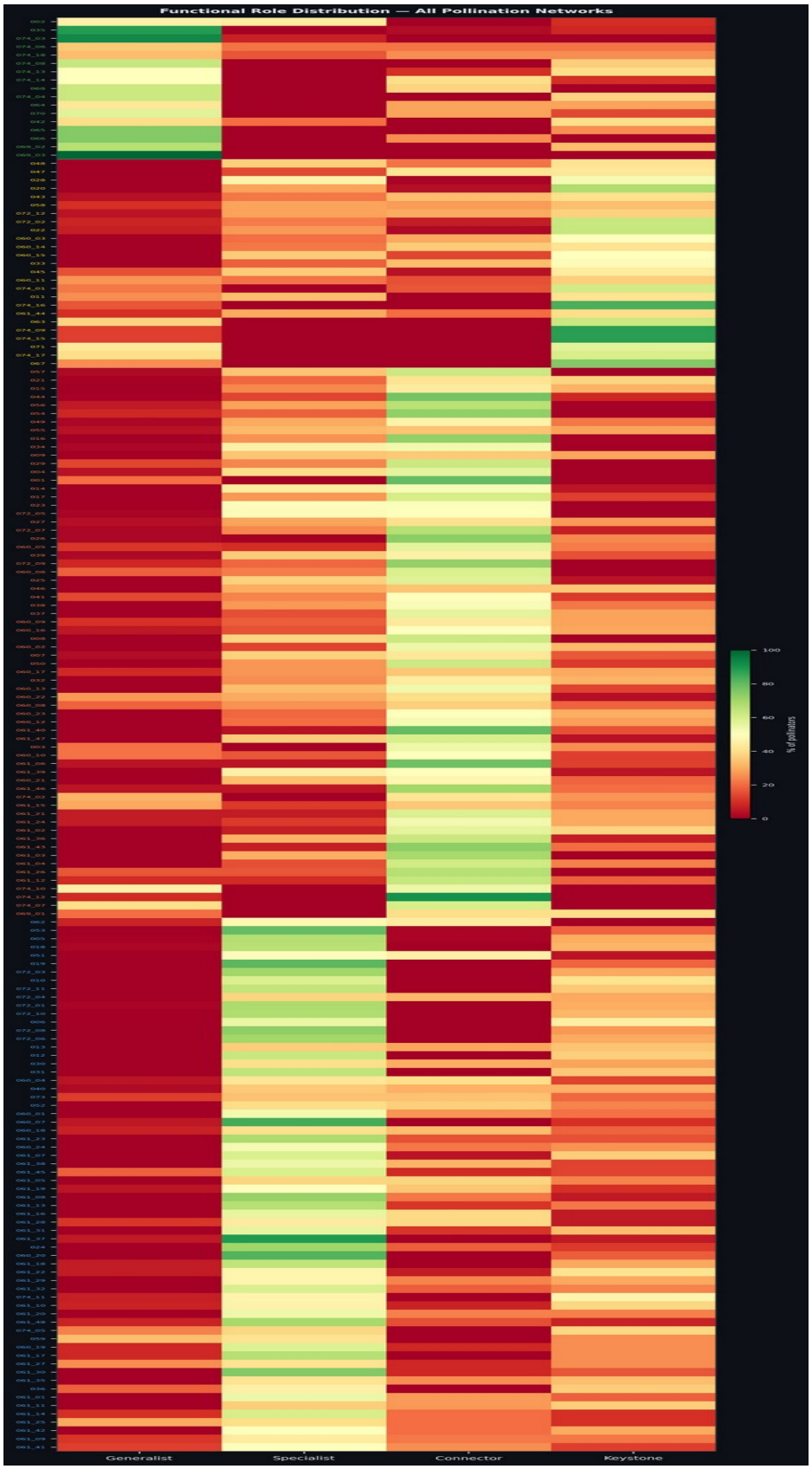

Functional Role Distribution — All Pollination Networks
Generalist
Specialist
Connector
Keystone
100
80
60
40
20
0
% of pollinators

Figure 15. Role Distribution Heatmap: *Functional role distribution heatmap for all 172 pollination networks (rows), showing the percentage of pollinators assigned to each of the four GNN roles (columns: Generalist, Specialist, Connector, Keystone). The heatmap illustrates the diversity of role compositions across the corpus and shows that Generalist dominance occurs in only a small subset of networks, often associated with higher connectance, although this pattern should be interpreted as a tendency rather than a strict rule.*

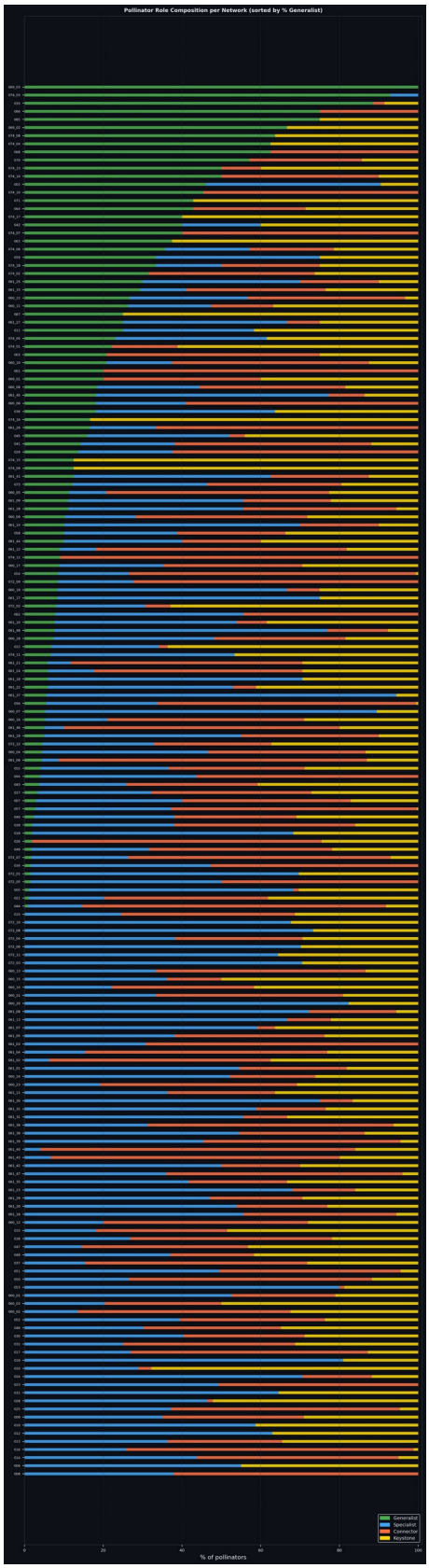

Pollinator Role Composition per Network (sorted by % Generalist)
Generalist
Specialist
Connector
Keystone
0
20
40
60
80
100
% of pollinators

Figure 16. Stacked Role Composition: *Pollinator role composition per network (stacked horizontal bars), sorted by descending % Generalist. Each bar represents the full 100% role composition (green: Generalist, blue: Specialist, orange: Connector, yellow: Keystone). The ordering reveals a gradual transition from a small number of Generalist-dominated networks at the top to a majority of Specialist- and Connector-dominated networks below, consistent with the connectance–generalism relationship observed in Figure 11.*

4.8 Species Transplantation Analysis

The transplantation of "Unidentified sp16 M_PL_050" into the target community (M_PL_050, 17 plants × 51 pollinators) produced the following network-level changes:

- Interactions increased from 129 to 132 (+3 new interactions), as the transplanted species formed connections with three target plants (Bituminaria bituminosa, Euphorbia obtusifolia, and Sideritis nervosa), all at moderate to high interaction strength.
- Connectance increased marginally from 0.1488 to 0.1493 (+0.0005), reflecting the dilution effect of adding one new node to an already relatively dense small network.
- Nestedness (NODF) decreased slightly from 38.94 to 38.06 (Δ = -0.88), suggesting that the transplanted species' interaction profile is less nested than the existing community average — likely because it forms interactions with moderately-connected (not maximally-connected) plants.
- Modularity (Q) decreased from 0.4627 to 0.4502 (Δ = -0.0125), indicating that the transplanted species bridges across existing modules by connecting with plants in different communities, slightly reducing overall community structure.
- LCC (Largest Connected Component) size increased by 1 (from 68 to 69 nodes), confirming successful network integration of the new species.

The delta centrality analysis reveals that the introduction of the transplanted species decreases the degree centrality and betweenness centrality of many existing pollinators by small but consistent amounts (~0.001-0.003 decrease), as the new species competes for pollination roles. The eigenvector centrality before-versus-after scatter plot shows that most species lie very close to the "no change" diagonal, confirming that the systemic influence of existing species is largely preserved despite the new introduction.

The Resilience Curve comparison (before vs. after transplant) shows marginal improvement in network resilience following transplantation: the after-transplant curve lies slightly above the before-transplant curve for most species removal fractions, consistent with the additional connectivity provided by the new interaction links. However, the improvement is small (within noise bounds), reflecting the limited structural impact of adding a single generalist pollinator to a 52-species community.

This result has an important conservation implication: the managed introduction of pollinators from structurally similar networks can modestly improve the resilience of target communities without destabilising existing interaction architecture. The key constraint is matching the interaction profile of the introduced species to the structural needs of the target network — in particular, the transplanted species should preferentially interact with bridge plants (high betweenness) rather than with already over-represented species.

Table 3. Network metric comparison before and after transplantation of ‘Unidentified sp16 M_PL_050’ into the target community (M_PL_050: 17 plants × 51 pollinators).

Metric	Before	After	Δ Change
Plants	17	17	0

Pollinators	51	52	+1
Interactions	129	132	+3
Connectance	0.1488	0.1493	+0.0005
Nestedness (NODF)	38.94	38.06	−0.88
Modularity (Q)	0.4627	0.4502	−0.0125
Communities	6	6	0
LCC Size	68	69	+1

Source: transplant_summary.csv. The transplanted species formed 3 new interactions with Bituminaria bituminosa, Euphorbia obtusifolia, and Sideritis nervosa at near-maximal interaction strength (1.0). Delta centrality metrics for all 51 existing pollinators are reported in transplant_centrality_report.csv.

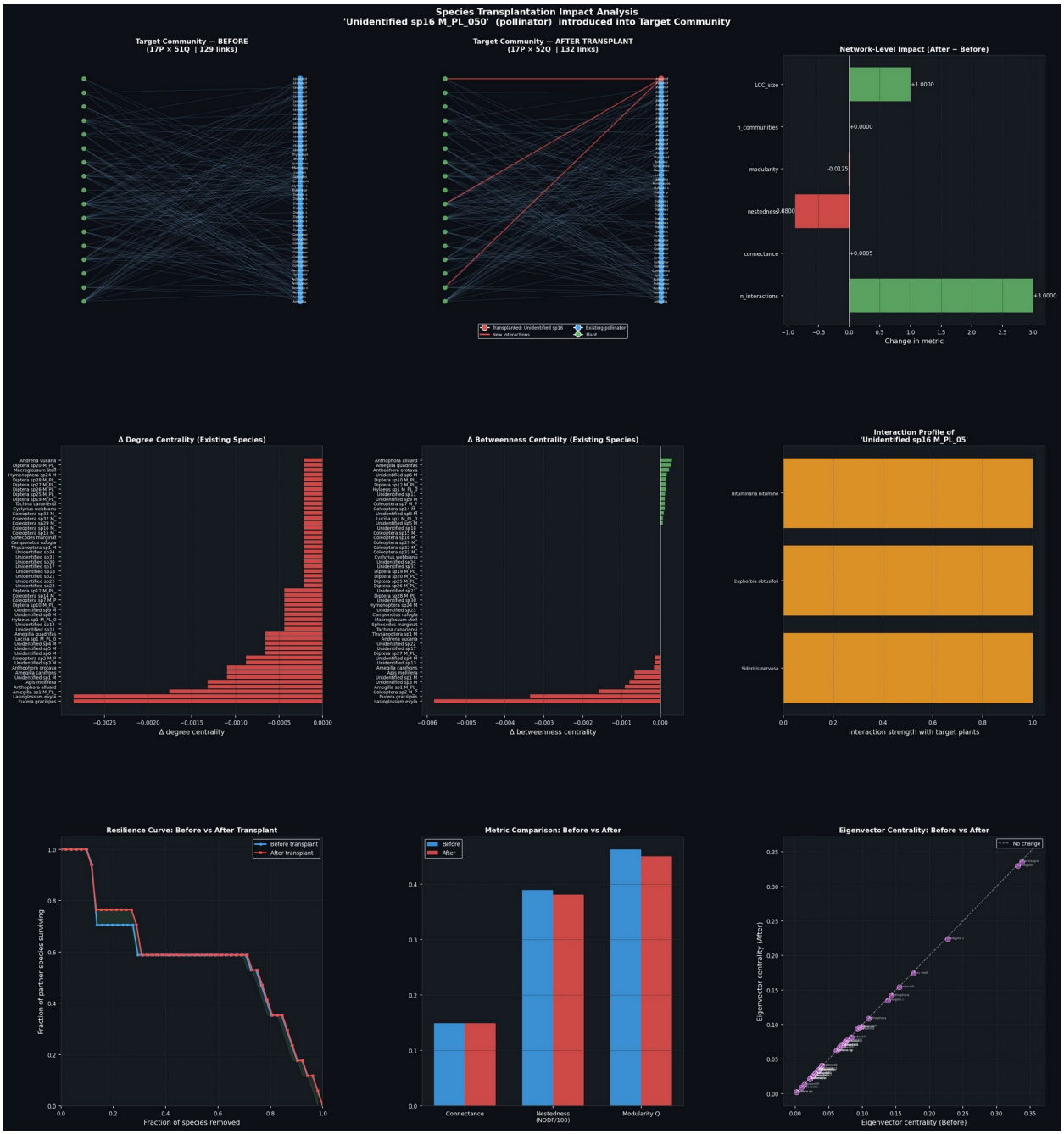


Figure 7. Species Transplantation Impact Analysis: introduction of 'Unidentified sp16 M_PL_050' (pollinator) into the M_PL_050 target community. Top row: bipartite network before (17P × 51 pollinators, 129 links) and after transplant (17P × 52 pollinators, 132 links), with new interactions shown in red, and network-level metric changes (bar chart). Middle row: Δ degree centrality and Δ betweenness centrality for all existing pollinators, and the transplanted species' interaction profile with its three new plant partners. Bottom row: resilience curves before vs. after transplant, before/after metric bar chart comparison, and eigenvector centrality scatter (before vs. after).

4.9 Significance of Analytical Figures

The following notes describe the specific analytical role of each figure or figure group in the report. They are intended to orient the reader navigating the document and to make explicit the evidential chain from raw network data to the conclusions drawn in Section 5.

Figures 1–6 — Continental Analysis Panels

Each figure is a nine-panel diagnostic for one continental pollination network. Together they make the core comparative argument of the report: no two continents share the same network architecture, and structure (nestedness versus modularity) directly predicts collapse speed, early warning window, and the magnitude of resilience gain from adaptive rewiring. The Africa panel (Figure 1) illustrates a high-nestedness, hub-driven collapse regime; the South America panel (Figure 6) illustrates the opposite extreme — extreme modularity protecting pollinators up to a high stress threshold but with asymmetric species sensitivity. Without these six figures, the continental comparisons in Section 4.1 remain abstract numbers.

Figure 7 — Species Transplantation Impact

This is the only figure in the report that shows the network-level effect of adding a species rather than removing one. The before/after bipartite panels, delta centrality charts, and resilience curve comparison together ground the conservation recommendation in Section 4.8: managed introductions can modestly improve resilience, but only when the transplanted species is structurally matched to bridge gaps rather than reinforce already well-connected nodes.

Figure 8 — GNN Quality Metric Distributions

This figure provides strong support for the reliability of the GNN analysis prior to interpreting role-based results. The AUC histogram shows that 171 of 172 networks exceed the acceptance threshold of 0.85 (mean = 0.960), indicating that the GraphSAGE model accurately captures network structure across the corpus. The silhouette histogram (mean = 0.384) further indicates that the four-role clustering is reasonably well-separated and internally cohesive. Together, these metrics justify proceeding with the interpretation of role proportions and correlations reported in Section 4.7.

Figures 9–10 — Summary Panel, Dominant Role Donut, Role % Boxplot

These figures establish the main corpus-level pattern: Connector and Specialist roles are nearly equally dominant across the 172 networks (38% vs. 37% of networks), while Generalist-dominant structures are comparatively rare (10%). The boxplot (Figure 10) adds an important nuance—high median proportions do not imply low variability. Specialist and Connector roles span a wide range (approximately 0–90%) across networks, indicating substantial heterogeneity in network organisation. The summary panel (Figure 9) integrates these distributions with quality and size-based patterns, providing a compact overview of the full analysis.

Figure 11 — Network Connectance vs. Role Proportions

This figure represents the quantitative core of the GNN findings. The four scatter plots show that Generalist proportion increases with connectance ($r = 0.48$), while Specialist ($r = -0.26$, $p = 0.001$) and Connector ($r = -0.20$, $p = 0.009$) proportions decrease. In contrast, Keystone proportion shows

no statistically significant relationship with connectance ($r = 0.07$, $p = 0.345$). This suggests that, unlike the other roles, Keystone prevalence is not strongly driven by interaction density and may instead depend on more complex structural characteristics captured in the learned embedding space.

Figures 12–13 — Role by Network Size, Size Bubble Chart

These figures demonstrate that network size influences role composition through mechanisms distinct from connectance alone. The size-binned bar chart (Figure 12) shows that Generalist proportion declines sharply from small to large networks (approximately 24% → 4%), while Specialist proportion increases, consistent with niche-partitioning theory. Connector proportion peaks in medium-sized networks and decreases slightly in larger systems, suggesting a redistribution of bridging roles. Keystone proportion remains relatively stable across size classes. The bubble chart (Figure 13) visualises the full network space, where Connector-dominated networks tend to occupy larger, species-rich regions, while Keystone-dominant networks appear more frequently among smaller, denser systems.

Figure 14 — GNN Quality Scatter (Silhouette vs. AUC)

This figure complements Figure 8 by examining model quality at the level of individual networks. Plotting silhouette score against AUC and colour-coding by dominant role shows no clear evidence that any role type is associated with systematically lower model performance. All four roles are represented within the high-AUC and acceptable-silhouette region, indicating that the four-role framework is consistently recoverable across networks of varying size and structure.

Figures 15–16 — Role Heatmap, Stacked Composition Bars

These figures provide the complete per-network evidence underlying the aggregate results. The heatmap (Figure 15) displays the full distribution of role proportions across all 172 networks, allowing identification of outliers and patterns not visible in summary statistics. The stacked bar plot (Figure 16), sorted by decreasing Generalist proportion, provides a clear visual representation of the connectance–generalism relationship: a transition from a small number of Generalist-dominated networks to a majority of Specialist- and Connector-dominated systems. This ordering makes the gradient directly interpretable without reliance on correlation values alone.

5. Conclusion and Future Aspects

This project has delivered a multi-scale, multi-methodological network science analysis of pollination system stability and collapse, integrating classical graph metrics, simulation frameworks, machine learning, and deep learning into a unified analytical pipeline. Our principal findings may be summarised as follows.

First, continental pollination networks exhibit vastly different structural signatures. South America's highly modular architecture (Q = 0.781) confers resistance to climate stress but makes it sensitive to module-level collapse. North America's integrated, low-modularity network (Q = 0.268) is more resilient to random extinction but more vulnerable to targeted removal of generalist species. Africa's highly nested structure (NODF = 18.2) enables it to tolerate high stress up to a critical threshold, after which collapse is rapid and synchronous — a pattern consistent with the tipping point dynamics described by Lever et al. (2014).

Second, our GNN analysis of 172 networks provides the largest and most systematic data-driven characterisation of pollinator functional roles to date. The discovery that Generalist proportion is strongly correlated with connectance (r = 0.48) while Keystone proportion is connectance-independent provides the first GNN-based quantitative support for classical ecological theory about disproportionate-impact species. The Connector-Specialist dichotomy (38% vs. 37% of dominant network types) implies that conservation interventions must be tailored to network architecture: bridge species in Connector networks, and specialist diversity in Specialist networks.

Third, eigenvector centrality consistently outperforms degree centrality and betweenness as a predictor of extinction risk across all datasets. This finding argues for the adoption of eigenvector-based prioritisation frameworks in conservation planning, as species with high eigenvector centrality exert systemic network influence that extends far beyond their direct interactions.

Fourth, adaptive rewiring provides meaningful resilience gains across all networks (particularly nested and smaller networks), providing computational support for conservation strategies that preserve or enhance interaction flexibility — for example, through habitat restoration that enables pollinators to expand their foraging ranges.

Fifth, early warning signals (rising variance and autocorrelation) are detectable at stress levels well before catastrophic collapse, suggesting that community biomass monitoring programs designed with network theory in mind could provide actionable warnings 10-30 years before irreversible ecosystem transitions.

Sixth, the species transplantation analysis demonstrates that managed introductions of ecologically compatible pollinators can modestly improve network resilience without disrupting existing interaction architecture,

provided the introduced species is selected to bridge structural gaps rather than reinforce already well-connected nodes.

-

6. References

Bascompte, J., & Scheffer, M. (2023). The resilience of plant-pollinator networks. Annual Review of Entomology, 68, 363-380. https://doi.org/10.1146/annurev-ento-120120-102424

Bascompte, J., Jordano, P., Melián, C. J., & Olesen, J. M. (2003). The nested assembly of plant-animal mutualistic networks. Proceedings of the National Academy of Sciences, 100(16), 9383-9387.

Bastolla, U., Fortuna, M. A., Pascual-Garcia, A., Ferrera, A., Luque, B., & Bascompte, J. (2009). The architecture of mutualistic networks minimizes competition and increases biodiversity. Nature, 458(7241), 1018-1020.

CaraDonna, P. J., Petry, W. K., Brennan, R. M., Cunningham, J. L., Bronstein, J. L., Waser, N. M., & Sanders, N. J. (2017). Interaction rewiring and the rapid turnover of plant-pollinator networks. Ecology Letters, 20, 385-394.

Cardinale, B. J. et al. (2012). Biodiversity loss and its impact on humanity. Nature, 486, 59-67.

Dakos, V., & Bascompte, J. (2014). Critical slowing down as early warning for the onset of collapse in mutualistic communities. Proceedings of the National Academy of Sciences, 111(49), 17546-17551.

Dakos, V., Scheffer, M., van Nes, E. H., Brovkin, V., Petoukhov, V., & Held, H. (2012). Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data. PLoS ONE, 7(7), e41010.

Dunne, J. A., Williams, R. J., & Martinez, N. D. (2002). Network structure and biodiversity loss in food webs: robustness increases with connectance. Ecology Letters, 5(4), 558-567.

Gilarranz, L. J., Rayfield, B., Linan-Cembrano, G., Bascompte, J., & Gonzalez, A. (2017). Effects of network modularity on the spread of perturbation impact in experimental metapopulations. Science, 357(6347), 199-201.

Goulson, D., Nicholls, E., Botías, C., & Rotheray, E. L. (2015). Bee declines driven by combined stress from parasites, pesticides, and lack of flowers. Science, 347(6229), 1255957.

Hamilton, W., Ying, Z., & Leskovec, J. (2017). Inductive representation learning on large graphs. Advances in Neural Information Processing Systems (NeurIPS), 30, 1024-1034.

Lever, J. J., Van Nes, E. H., Scheffer, M., & Bascompte, J. (2014). The sudden collapse of pollinator communities. Ecology Letters, 17(3), 350-359.

Memmott, J., Waser, N. M., & Price, M. V. (2004). Tolerance of pollination networks to species extinctions. Proceedings of the Royal Society B, 271(1557), 2605-2611.

Ollerton, J., Winfree, R., & Tarrant, S. (2011). How many flowering plants are pollinated by animals? Oikos, 120, 321-326.

Rezende, E. L., Lavabre, J. E., Guimarães, P. R., Jordano, P., & Bascompte, J. (2007). Non-random coextinctions in phylogenetically structured mutualistic networks. Nature, 448, 925-928.

Rohr, R. P., Saavedra, S., & Bascompte, J. (2014). On the structural stability of mutualistic systems. Science, 345(6195), 1253497.

Scheffer, M., Bascompte, J., Brock, W. A., Brovkin, V., Carpenter, S. R., et al. (2009). Early-warning signals for critical transitions. Nature, 461(7260), 53-59.

Simpson, D. T., Weinman, L. R., Genung, M. A., Roswell, M., MacLeod, M., & Winfree, R. (2022). Many bee species, including rare species, are important for function of entire plant-pollinator networks. Proceedings of the Royal Society B, 289, 20212689.

Tilman, D., Isbell, F., & Cowles, J. M. (2014). Biodiversity and ecosystem functioning. Annual Review of Ecology, Evolution, and Systematics, 45, 471-493.

Valdovinos, F. S., de Espanés, P. M., Flores, J. D., & Ramos-Jiliberto, R. (2013). Adaptive foraging allows the maintenance of biodiversity of pollination networks. Oikos, 122, 907-917.

Vieira, M. C., & Almeida-Neto, M. (2015). A simple stochastic model for complex coextinctions in mutualistic networks: robustness decreases with connectance. Ecology Letters, 18(2), 144-152.

Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., & Philip, S. Y. (2020). A comprehensive study on graph neural networks. IEEE Transactions on Neural Networks and Learning Systems, 32(1), 4-24.